
AT-Dec-POSG: Formalizing and Solving Adversarial Team Games with Decentralized Partial Observability

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Abstract

1 Adversarial team problems such as Doudizhu and multi-contract futures execution
2 share a specific information bottleneck: teammates must cooperate against external
3 pressure under partial observability, yet execution-time private state cannot be
4 shared. We formalize this regime as *Adversarial Team Decentralized Partially*
5 *Observable Stochastic Games* (AT-Dec-POSG), a constrained subclass of partially
6 observable adversarial team games defined by decentralized partial observability
7 (C3) and strict intra-team information isolation (C4). We establish that C4 can
8 induce strictly positive coordination loss, that the team-side control problem re-
9 duces to public-belief control once the adversary is fixed, and that coarse legal
10 information leaves an irreducible coordination residue. These structural results
11 motivate a C4-compliant residual-reduction framework in which a *Nested Belief*
12 *Critic*, an *Implicit Action-Encoded Coordination* mechanism, and a *Structured*
13 *League Robustness Curriculum* with a *Team Minimax Exploiter* reduce the belief,
14 legal-coordination, and adversary-coverage residuals, respectively. On Doudizhu,
15 against the PerfectDou baseline under a bidding-enabled protocol, our full system
16 reaches a bidirectional average win percentage (WP) of 0.5152 and average deal
17 points (ADP) of 0.3183. On a replay-based structural stress test derived from
18 multi-contract futures execution, matched-budget comparisons show nonzero co-
19 ordination recovery, while follow-up diagnostics expose a completion/robustness
20 trade-off under public-trace interventions. We propose a C4-compliant residual-
21 reduction framework for AT-Dec-POSG, supported by structural guarantees and
22 validated by consistent empirical gains over strong baselines across two structurally
23 aligned domains.

24 1 Introduction

25 Many adversarial team problems are difficult for a reason that standard MARL abstractions do not
26 isolate cleanly: teammates face external pressure while each agent sees only a private slice of state and
27 cannot inspect teammate-private information at execution time. This C4-style information constraint
28 is practically important and underemphasized: once enforced, the public interaction trace becomes
29 the only legal coordination channel. Doudizhu and replay-based multi-contract futures execution
30 exhibit exactly this bottleneck.

31 Standard formulations each capture only part of this regime. Cooperative CTDE methods such
32 as QMIX, MAPPO, and HAPPO assume a shared team objective but no explicit adversarial side
33 [12, 17, 26]. Mixed cooperative-competitive methods such as MADDPG use centralized critics with
34 decentralized actors under strategic opposition, but they do not isolate C4-style strict intra-team
35 execution-time information restrictions and the resulting coordination loss as first-class objects
36 [14]. Communication-oriented MARL relies on explicit teammate channels that are illegal when
37 private state must remain hidden [5, 6]. ATMGs are the closest precursor because they capture team-
38 versus-adversary structure, but they do not elevate strict intra-team isolation and public-action-only

39 coordination to first-class execution constraints [11]. The gap we study is therefore not “beyond
40 POSGs” in generality, but “within POSGs” in information structure.

41 The operational bottleneck is therefore a policy-class restriction, not missing game-theoretic general-
42 ity. Once teammates are limited by **C3**, decentralized partial observability, and **C4**, strict intra-team
43 execution-time information isolation, the team must coordinate against an adversary while condi-
44 tioning only on local action-observation history and the public interaction trace. The public action
45 history becomes the only legal coordination medium, and the admissible decentralized team class
46 is strictly smaller than the centralized one. We formalize exactly this regime as *Adversarial Team*
47 *Decentralized Partially Observable Stochastic Games* (AT-Dec-POSG), best read as a constrained
48 lens on this overlooked information structure rather than as a claim to solve adversarial team games
49 in full generality.

50 This structure creates three coupled learning problems: noisy value estimation under C3, legal
51 coordination under C4, and cycling in adversarial team training. Accordingly, the paper is not
52 organized around three unrelated modules, but around a legality-first coordination framework: a
53 *Nested Belief Critic*, an *Implicit Action-Encoded Coordination* mechanism, and a *Structured League*
54 *Robustness Curriculum* with a *Team Minimax Exploiter* reduce the residuals created by this C3/C4-
55 constrained setting.

56 Doudizhu is the primary domain because it exposes this structure cleanly, while replay-based multi-
57 contract futures execution serves as a narrower structural stress test. The goal is not to compare
58 WP with shortfall numerically, but to test whether the same public-channel recovery pattern and
59 intervention sensitivity reappear across two structurally aligned domains.

60 Our contributions are three-fold:

- 61 1. **C4-constrained adversarial-team lens.** We isolate a practically important but underemphasized
62 information regime in adversarial team problems, formalize it as AT-Dec-POSG, prove mono-
63 tonicity under C4 restriction, and show via a strict-gap construction that C4 can induce a positive
64 coordination barrier.
- 65 2. **Legal public-trace coordination framework.** We give a C4-compliant residual-reduction frame-
66 work in which a Nested Belief Critic handles critic-side hidden-state estimation, a public-trace
67 coordination mechanism exploits the only legal teammate-side channel, and a league-side robust-
68 ness curriculum with a Team Minimax Exploiter improves adversary coverage without changing
69 execution-time legality.
- 70 3. **Structural theory and cross-domain evidence.** We show that, once the adversary is fixed, the
71 team-side control problem reduces to public-belief control, and that overly coarse teammate-side
72 legal information leaves an information-theoretically irreducible coordination residue. Across
73 Doudizhu and a replay-based execution stress test, the same intervention and recovery pattern
74 reappears, supporting a shared public-trace coordination mechanism; the appendix collects the
75 remaining propositions as auxiliary and supporting results.

76 Empirically, the paper tests a structural prediction rather than a loose transfer story: across both
77 domains, the full system should beat the base, recover part of the coordination-loss gap through
78 Module B, depend on teammate-originating public traces, and concentrate information on the most
79 committed public actions.

80 2 Related Work

81 **Cooperative and adversarial MARL.** The ambient setting is decentralized partial observability,
82 whose control complexity is already known to be severe in Dec-POMDP-style formulations [2, 16].
83 On the cooperative side, CTDE and value-factorization methods such as QMIX, MAPPO, and HAPPO
84 stabilize decentralized execution with richer training-time information [12, 17, 26]. On the adversarial
85 side, MADDPG and later game-theoretic MARL formulations treat mixed cooperative-competitive
86 interaction and equilibrium learning more explicitly, often again through centralized training signals
87 paired with decentralized execution [11, 13, 14, 18]. What remains unisolated in these lines is the
88 exact information structure central to our paper: team-vs-adversary interaction under decentralized
89 partial observability together with strict intra-team execution-time isolation and an explicit C4-style
90 legality constraint on coordination.

Implicit communication. Communication-based MARL has shown that learned message passing can dramatically improve coordination under partial observability. DIAL/RIAL, CommNet, and TarMAC all rely on an explicit or architecturally privileged execution-time channel between teammates [5, 6, 19]. A closer line for our purposes is implicit communication through public actions: Bayesian Action Decoder (BAD), Simplified Action Decoder (SAD), and Learning to Communicate Implicitly by Actions all exploit observable actions to reveal private state without an explicit execution-time message channel [7, 9, 20]. Action-graph policies are related in spirit because they model inter-agent action dependencies, and stigmergic coordination methods likewise study indirect coordination through environmental traces [8, 22]. But these lines are developed primarily in cooperative Dec-POMDP settings and do not impose C4 as an execution-time legality constraint inside an adversarial team game. Our claim is narrower: Module B is a C4-compliant coordination mechanism whose only legal execution-time medium is the public action history.

Strategic self-play and Doudizhu AI. League-based self-play and opponent-aware training are central tools for large-scale strategic learning [21], and Minimax Exploiter is the most direct algorithmic predecessor for our Module C because it sharpens best-response pressure through opponent-value shaping [1]. In imperfect-information adversarial games, public-belief search methods such as ReBeL are also close in spirit because they elevate common/public belief to the control state and combine it with function approximation [4]; our setting differs in that the key extra restriction is the team-side C4 legality constraint, so the question is not only how to search over public beliefs, but how to coordinate legally when teammate-private state cannot be shared at execution. On Doudizhu specifically, prior systems span search-heavy and learned approaches including DeltaDou, Combinational Q-Learning, RLCARD-based evaluation pipelines, tree-search/action-prediction hybrids, DouZero, DouZero+, and PerfectDou [10, 23, 24, 27–30]. These works establish strong domain baselines and, in PerfectDou, show that privileged-value support can materially improve imperfect-information play. Our claim is narrower and more structural: the missing abstraction is not another stronger Doudizhu bot in isolation, but an information-structured adversarial team formulation that separates critic support, legal public-channel coordination, and league-side robustness as distinct sources of error.

3 Problem Formulation and Framework

AT-Dec-POSG definition. Let $\mathcal{M}$ denote the team and $\mathcal{K}$ the adversarial side. For agent i , let x_t^i denote its local non-public information state and let $\tau_t^i = (h_t^{\text{pub}}, x_t^i)$ be its full legal execution-time information, with h_t^{pub} the legal public interaction history. We write $\Pi_{\mathcal{M}}$ for the unconstrained team policy class and $\Pi_{\mathcal{M}}^{\text{Dec}} \subseteq \Pi_{\mathcal{M}}$ for the subset that satisfies C4.

Definition 1 (AT-Dec-POSG). *An Adversarial Team Decentralized Partially Observable Stochastic Game is a partially observable adversarial team game with: **C1**, an explicit team-versus-adversary partition $(\mathcal{M}, \mathcal{K})$; **C2**, a zero-sum team-versus-adversary payoff structure summarized by a shared team-side scalar reward r_t , so that teammates optimize the same return while the adversarial side is evaluated in opposing payoff coordinates; **C3**, decentralized partial observability; and **C4**, strict intra-team execution-time isolation, meaning teammate i may condition only on $\tau_t^i = (h_t^{\text{pub}}, x_t^i)$, never on teammate-private state or a private execution-time message.*

Accordingly, we write the team-side scalar reward as r_t : for teammates $i \in \mathcal{M}$, $r_t^i = r_t$, while for adversarial agents $k \in \mathcal{K}$, $r_t^k = -r_t$.

The resulting decentralized zero-sum minimax objective is

$$V^*(s) = \sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}} \inf_{\pi_{\mathcal{K}}} V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}(s).$$

Lemma 1 states the C4 barrier, while Examples 1 and 2 show it in a native card-game domain and a narrower replay-based execution stress test.

Lemma 1 (Monotonicity Under C4 Restriction). *For every initial state s ,*

$$\sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}} \inf_{\pi_{\mathcal{K}}} V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}(s) \leq \sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}} \inf_{\pi_{\mathcal{K}}} V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}(s).$$

When the centralized optimum is attained, equality requires that an optimal centralized team policy admit a decentralized realization that depends on no teammate-private observations beyond the public history.

Domain instantiation. Lemma 1 is the barrier statement: once C4 removes teammate-private execution-time dependence, it can remove attainable value even when teammates share reward. The point of the formulation is therefore not to claim a broader adversarial-team game class, but to isolate the overlooked case in which the legal public trace is the only remaining coordination channel. The two domains are paired because they share exactly this information structure: the team faces external pressure, each agent sees only local/private state, and legal coordination must flow through a public action trace. Doudizhu is the cleaner primary domain; the replay-based execution benchmark is the narrower stress test. Additional tuple-level background, the strict-gap bit-game example, the conservative upper envelope, and domain-instantiation details appear in Appendix A.4, Appendix A.2, Appendix A.3, Appendix A.5, and Appendix A.17.

Example 1 (Doudizhu as a role-assignment AT-Dec-POSG continuation game). *Doudizhu is naturally modeled as a public role-assignment continuation game. The public bidding stage first generates a bid history h^{bid} , after which the game rule deterministically assigns the landlord role L . The continuation AT-Dec-POSG is then defined with $\mathcal{K} = \{L\}$, $\mathcal{M} = \mathcal{N} \setminus \{L\}$, and farmer-side local observation equal to private hand plus inherited public bid/play history. All AT-Dec-POSG structural results are applied to the continuation game after public role assignment. The bidding history is treated as legal public pre-history inherited by the continuation game. Appendix A.5 gives the full formalization.*

Example 2 (Replay-based multi-contract futures execution as a structural AT-Dec-POSG stress test). *Under the task-level adversary abstraction used here, the replay-based multi-contract futures execution benchmark exhibits the same C3/C4 pattern: each execution agent sees only contract-local book and inventory state plus the public tape, the team reward is shared negative aggregate execution cost, and teammate interaction can occur only through publicly visible order flow. Because the underlying proprietary market data and held-asset details cannot be directly disclosed, the second domain is instantiated as a replay-based simulated environment that preserves the same public-trace structure without exposing raw internal trading records. The market side is represented by aggregate liquidity response in the replayed stream rather than by an explicit learned opponent.*

Example 1 is the cleaner anchor because the public channel is already part of Doudizhu’s legal execution semantics through inherited bids and plays. Example 2 is intentionally narrower than Example 1: it is a structural stress test under adversarial scenario families rather than a claim about fully explicit online opponent learning. Together, Lemma 1 and Examples 1–2 set up the later structural theory: C4 creates the barrier, the two domains expose the legal public trace through which that barrier can only be partially repaired, Proposition 1 identifies the right reduced control state, and Proposition 3 lower-bounds what remains unrecoverable; the compact structural mapping table for the two domains is reported in Appendix A.7.

4 Method

AT-Dec-POSG suggests a residual-reduction view rather than a collection of independent architectural tricks. Its error naturally decomposes into a belief residual ε_A under C3, a legal-coordination residual ε_B under C4, and an adversary-coverage residual ε_C under adversarial team training:

$$\varepsilon = \max\{\varepsilon_A + \varepsilon_B, \varepsilon_C\} \leq 2\varepsilon_A + \varepsilon_B + \varepsilon_C \leq \underbrace{\sqrt{2\lambda_1 L_{\text{distill}}^*}}_{\text{Module A}} + \underbrace{\max(0, \Delta_{\text{coord}} - \delta_G T)}_{\text{Module B}} + \underbrace{\frac{\text{diam}(V)}{|\Pi_{\mathcal{K}}^L|}}_{\text{Module C}}.$$

Here L_{distill}^* is the final measured distillation loss, λ_1 is the task-dependent conversion constant from that loss scale into value units, Δ_{coord} is the total coordination gap between the explicit-latent and no-coordination protocols, δ_G is the assumed per-step recoverable coordination gain across the T informative decision steps, and $\text{diam}(V) = V_{\text{max}} - V_{\text{min}}$ together with $|\Pi_{\mathcal{K}}^L|$ gives the value range divided by the size of the adversary-side league pool. The calibrated right-hand side is the closed-form reading of Proposition 10 from Appendix A.16. Modules A, B, and C are therefore residual reducers rather than independent add-ons: training-only belief support shrinks ε_A , informative legal public actions shrink ε_B , and finite-league exploiter pressure shrinks ε_C . All privileged information and auxiliary labels remain training-only, while deployed actors stay measurable with respect to $\tau_t^i = (h_t^{\text{pub}}, x_t^i)$. We keep each domain’s base learner fixed: DMC in Doudizhu and PPO in replay-based execution; implementation details are deferred to Appendix A.25, Appendix A.26, and Appendix A.27.

190 **Module A: Nested Belief Critic.** Under C3, an actor conditioned only on τ_t^i faces uncertainty
 191 about hidden strategic state. We therefore use a CTDE critic with input

$$x_t^{\text{critic},i} = o_t^i \oplus s_t^{\text{full}} \oplus \hat{b}_t^{-i},$$

192 where $x_t^{\text{critic},i}$ is the critic feature vector for agent i , o_t^i is its current local observation, $\oplus$ denotes
 193 feature concatenation, s_t^{full} is privileged training-time state, and $\hat{b}_t^{-i}$ is a belief over unobserved
 194 other-player state. The loss is

$$\mathcal{L}_A = \mathcal{L}_{\text{TD}} + \lambda_{\text{dist}} \mathcal{L}_{\text{distill}} + \lambda_{\text{belief}} \mathcal{L}_{\text{belief}}.$$

195 Here $\mathcal{L}_{\text{TD}}$ is the base temporal-difference or actor-critic value loss, $\mathcal{L}_{\text{distill}}$ is the teacher-to-student
 196 distillation term, $\mathcal{L}_{\text{belief}}$ is the hidden-state prediction auxiliary, and $\lambda_{\text{dist}}, \lambda_{\text{belief}} > 0$ weight the two
 197 auxiliaries. The three terms play distinct roles. $\mathcal{L}_{\text{TD}}$ remains the task loss, $\mathcal{L}_{\text{distill}}$ encourages the
 198 decentralized critic to inherit structure from a richer privileged teacher, and $\mathcal{L}_{\text{belief}}$ stabilizes value
 199 prediction by making hidden opponent-side state partially predictable from the local history. The
 200 key design choice is that privileged features improve value learning but never enter the execution-
 201 time actor path. In the PPO-style execution benchmark, actor advantages are computed from the
 202 decentralized student value path; the privileged teacher branch is training-only and enters only through
 203 $\mathcal{L}_{\text{distill}}$. In the paper’s residual language, Module A is meant to shrink the belief residual ε_A rather
 204 than to change the legality of execution itself. Module-A architecture details and belief targets are
 205 given in Appendix A.25.

206 **Module B: Implicit Action-Encoded Coordination.** The core legality condition is that execution-
 207 time policies keep the same signature

$$a_t^i \sim \pi^i(\cdot \mid \tau_t^i).$$

208 Here a_t^i is agent i ’s legal execution-time action and $\pi^i(\cdot \mid \tau_t^i)$ is its decentralized stochastic policy
 209 conditioned only on legal local information. No teammate-private latent tensor, message vector, or
 210 privileged state enters the actor path at inference time.

211 Training then adds sender/receiver auxiliaries that make legal public actions more informative:

$$\begin{aligned} \mathcal{L}_{\text{sender}} &= \text{CE}(\text{coord_sender_logits}, y_{\text{sender}}^i), \\ \mathcal{L}_{\text{receiver}} &= \text{CE}(\text{coord_receiver_logits}, y_{\text{receiver}}^{i \rightarrow j}). \end{aligned}$$

212 In both expressions, CE denotes cross-entropy, coord_sender_logits and coord_receiver_logits are
 213 the training-time sender/receiver classifier heads, y_{sender}^i is agent i ’s sender-side coordination label,
 214 and $y_{\text{receiver}}^{i \rightarrow j}$ is teammate j ’s reconstruction target for information revealed by sender i ’s public action
 215 trace.

216 **Claim 1** (C4 legality of Module B). *If each deployed actor conditions only on $\tau_t^i = (h_t^{\text{pub}}, x_t^i)$, all*
 217 *sender/receiver auxiliaries are used only through training losses, and no auxiliary logit, target, latent,*
 218 *or privileged feature is routed into the execution-time actor, then Module B preserves C4.*

219 The proof is the measurability argument in Proposition 5: Module B may bias training toward more
 220 informative public actions, but the deployed policy still depends only on local and public history.
 221 Proposition 1, Proposition 2, and Proposition 3 explain why public-history informativeness matters
 222 structurally and what remains impossible when that legal information is still too weak. Additional
 223 bidding-stage discussion, the occupancy-average public-channel action-information proposition,
 224 the full sender/receiver implementation text, and the intervention protocol are in Appendix A.15,
 225 Appendix A.26, and Appendix A.19.

226 The sender/receiver losses are domain instantiations of the same structural idea. In Doudizhu, y_{sender}^i
 227 summarizes private hand-strength / remaining-hand categories and $y_{\text{receiver}}^{i \rightarrow j}$ asks whether a teammate
 228 can recover that summary from the bid/play trace. In multi-contract execution, let

$$u_t^i = \text{clip}\left(\frac{|r_t^i|}{\max(|r_0^i|, \epsilon_r)} \cdot \frac{T}{\max(T - t, 1)}, 0, 1\right), \quad \epsilon_r > 0,$$

229 denote the sender’s normalized urgency from remaining inventory and time-to-go. Here r_t^i is agent
 230 i ’s signed remaining inventory at step t , r_0^i is its initial target inventory, T is the episode horizon

in decision steps, ϵ_r is a small positive stabilizer, $\text{clip}(\cdot, 0, 1)$ truncates to $[0, 1]$, and $\text{bin}(\cdot)$ maps a scalar urgency into a discrete coordination label. We then use $y_{\text{sender}}^i = \text{bin}(u_t^i)$, while each receiver predicts the discretized teammate-mean urgency $y_{\text{receiver}}^j = \text{bin}(\frac{1}{N-1} \sum_{i \neq j} u_t^i)$, where N is the team size, from public tape and local history alone. These auxiliaries provide dense representation-learning signals; the actual preference for informative legal actions is still mediated by the shared RL objective. The claim is not that these auxiliaries directly estimate $I(a_t^i; o_{t+1}^j)$, but that a legal public action can carry recoverable teammate-relevant information under C4. Module B is the training-time mechanism used to make that possibility operational. In the residual decomposition, Module B is the component that directly targets ε_B , the legal-coordination residual.

Module C: Structured League Robustness Curriculum. Adversarial team training is prone to cycling, so Module C maintains three policy populations: main policies for robustness, snapshot policies for diversity, and exploiter policies for approximate best response. In the replay-based execution benchmark, this module is instantiated as a finite scenario-league robustness curriculum rather than as online opponent learning: a main balanced regime, frozen snapshot regimes, and a stressed-liquidity scenario family are sampled with main/snapshot/exploiter probabilities during rollout collection. In Doudizhu, the extra issue relative to two-player Minimax Exploiter is that the landlord faces a decentralized two-farmer team rather than a single opponent. We therefore use a symmetry-preserving joint farmer-side surrogate and write the exploiter rewards as

$$R_t^{\text{land-exp}} = R_t^{\text{env}} + B_t^{\text{land-exp}},$$

$$R_t^{\text{farm-exp}} = R_t^{\text{env}} + B_t^{\text{farm-exp}},$$

with shifted bonuses

$$B_t^{\text{land-exp}} = -\alpha\gamma(1 - d_t) \bar{V}_{\text{farmer-joint}}(s_{t+1}), \quad B_t^{\text{farm-exp}} = -\alpha\gamma(1 - d_t) \bar{V}_L(s_{t+1}),$$

where R_t^{env} is the original environment reward, $R_t^{\text{land-exp}}$ and $R_t^{\text{farm-exp}}$ are the exploiter-side shaped rewards, $B_t^{\text{land-exp}}$ and $B_t^{\text{farm-exp}}$ are the corresponding shaping bonuses, $\alpha > 0$ is the shaping strength, $\gamma \in (0, 1)$ is the discount factor, $d_t \in \{0, 1\}$ is the episode-termination indicator, and $\bar{V}_{\text{farmer-joint}}, \bar{V}_L$ are the nonnegative shifted surrogate values defined in Appendix A.23. To prevent reward-hacking cycles, the shifted shaping bonuses are constrained to be non-positive; the short proof is in Appendix A.23. We use the symmetry-preserving mean surrogate in the main system, and additional league details together with the min/mean/max surrogate comparison are reported in Appendix A.27 and Appendix A.22.

At the robustness end, we make only a fixed-adversary-pool claim: broader team-side finite-league coverage should not hurt the restricted minimax value. The formal statement is deferred to Appendix A.9. In the asymmetric 2-vs-1 Doudizhu case, we do not claim that the mean surrogate is an exact game-theoretic value identity; it is a symmetry-preserving finite-league proxy that keeps the landlord-side exploiter objective well-posed without introducing a centralized farmer controller on the execution path. The empirical comparison against alternative min/mean/max surrogates is therefore treated as supporting evidence rather than as a theorem; the summary appears in Section 6, and the compact numerical table is retained in Appendix A.31. In the residual decomposition, Module C is the component that targets ε_C , the adversary-coverage residual.

5 Structural Theory

Lemma 1 shows that C4 creates a genuine coordination barrier by shrinking the admissible team policy class. The three main results in this section sharpen that observation: public belief is the right reduced team-side state under C4 [15]; richer legal public traces are weakly better than coarser ones in the Blackwell sense [3]; and if legal teammate-side information remains too coarse, part of the coordination residue is information-theoretically irreducible. Appendix A.8 and Appendix A.9 record two supporting guarantees used by Modules B and C. Throughout the information-theoretic statements here and in the appendix, entropies and mutual informations are measured in bits.

Proposition 1 (Common-public-belief sufficiency under C4). *Consider a finite-horizon finite-state/action AT-Dec-POSG and fix a Markov adversary policy $\pi_K(a_t^K | s_t, h_t^{\text{pub}})$ with no additional hidden memory beyond the current latent/public state. Let each teammate history decompose as*

279 $\tau_t^i = (h_t^{\text{pub}}, x_t^i)$, where h_t^{pub} is the public history and x_t^i is teammate i 's private local information.
 280 Define the common public belief

$$b_t := \mathbb{P}(s_t, x_t^1, \dots, x_t^{|\mathcal{M}|} \mid h_t^{\text{pub}}).$$

281 Then the team best-response problem against $\pi_{\mathcal{K}}$ admits an equivalent coordinator formulation on
 282 state b_t : there exists an optimal prescription process $\Gamma_t = (\gamma_t^i)_{i \in \mathcal{M}}$, measurable with respect to b_t
 283 alone, where each γ_t^i maps x_t^i to a distribution over actions and each decentralized action can be
 284 implemented as

$$a_t^i \sim \gamma_t^i(\cdot \mid x_t^i), \quad i \in \mathcal{M}.$$

285 Appendix A.11 gives a full finite-space proof. The technical novelty is not the common-information
 286 reduction itself, but its use as the canonical best-response state after fixing the adversary under the C4
 287 legality constraint. Architecturally, Module A approximates the hidden-state part of this public-belief
 288 state, while Module B changes how informative the public-history update is for that belief. The
 289 formal legality statement for Module B is deferred to Appendix A.8; with legality fixed, the next
 290 proposition asks whether one legal public signal is structurally better than another.

291 **Proposition 2** (Blackwell monotonicity of legal public signals). *Fix an AT-Dec-POSG, a fixed*
 292 *adversary policy $\pi_{\mathcal{K}}$, and two C4-legal public-signal processes Y' and Y . Let $V_{\text{Dec}}^*(Y'; \pi_{\mathcal{K}})$ and*
 293 *$V_{\text{Dec}}^*(Y; \pi_{\mathcal{K}})$ denote the optimal decentralized team best-response values when policies may condition*
 294 *on $(x_t^i, Y'_{0:t})$ or $(x_t^i, Y_{0:t})$, respectively. If Y is a public garbling of Y' in the Blackwell sense, with*
 295 *any extra garbling randomness publicly shared, then*

$$V_{\text{Dec}}^*(Y'; \pi_{\mathcal{K}}) \geq V_{\text{Dec}}^*(Y; \pi_{\mathcal{K}}).$$

296 Appendix A.12 gives a full finite-space proof. Its role is to separate legality from usefulness: Module
 297 B does not create a new channel, but it does try to steer legal public actions toward a more informative
 298 legal signal structure. To quantify that induced informativeness, fix a coordination-critical event C_t ,
 299 a sender-private summary Z_t , and receiver-local information x_t^j , and define the legal coordination
 300 capacity

$$C_{\text{legal}} := I(Z_t; h_t^{\text{pub}} \mid x_t^j, C_t).$$

301 If $q_{\psi}(z \mid x_t^j, h_t^{\text{pub}}, C_t)$ is a held-out receiver probe with cross-entropy $\mathcal{L}_{\text{rec}}^{\text{held}} := \mathbb{E}[-\log_2 q_{\psi}(Z_t \mid$
 302 $x_t^j, h_t^{\text{pub}}, C_t)]$, then

$$C_{\text{legal}} \geq H(Z_t \mid x_t^j, C_t) - \mathcal{L}_{\text{rec}}^{\text{held}}.$$

303 Appendix A.14, Proposition 7, derives this lower bound and the matching no-public control probe
 304 used in the experiments.

305 **Proposition 3** (Irreducible legal-coordination residue). *Fix a C4-compliant team policy π , a fixed*
 306 *adversary policy, and the induced trajectory law $\mathbb{P}^{\pi}$. Let C_t be a coordination-critical event with*
 307 *occupancy mass $\rho_{\text{crit}}^{\pi} := \mathbb{P}^{\pi}(C_t) > 0$. On C_t , let $Z_t \in \mathcal{Z}$, $|\mathcal{Z}| \geq 2$, be an action-relevant sender-*
 308 *private summary indexing the continuation-equivalence class of the centralized prescription, so*
 309 *summary values that induce the same centralized continuation are merged. Let teammate j 's legal*
 310 *information be $L_t^j := \tau_t^j$. Assume that the continuation chosen by π induces a legal decoder $\hat{Z}_{\pi}(L_t^j)$*
 311 *and a nonnegative continuation-value gap D_{π} satisfying the margin condition*

$$D_{\pi} \geq \lambda \mathbf{1}\{\hat{Z}_{\pi}(L_t^j) \neq Z_t\} \quad \text{on } C_t$$

312 for some $\lambda > 0$. Write

$$\varepsilon_{\text{coord}}^{\pi} := \mathbb{E}^{\pi}[D_{\pi} \mathbf{1}_{C_t}]$$

313 for the coordination residue on that event. If

$$P_e^{\pi} := \inf_{\hat{Z}} \mathbb{P}^{\pi}(\hat{Z}(L_t^j) \neq Z_t \mid C_t),$$

314 where the infimum ranges over all measurable legal decoders $\hat{Z} : L_t^j \mapsto \mathcal{Z}$, and all H_{π} and I_{π} terms
 315 below are computed under $\mathbb{P}^{\pi}$, then

$$\varepsilon_{\text{coord}}^{\pi} \geq \lambda \rho_{\text{crit}}^{\pi} P_e^{\pi} \geq \lambda \rho_{\text{crit}}^{\pi} \left[\frac{H_{\pi}(Z_t \mid C_t) - I_{\pi}(Z_t; L_t^j \mid C_t) - 1}{\log_2 |\mathcal{Z}|} \right]_{+}.$$

Table 1: Doudizhu 10k-deck tournament results. Entries report seat-averaged (WP, ADP) for row method A against column method B ; larger values favor A .

$A \setminus B$	AT-Dec-POSG		PerfectDou		DouZero-WP		DouZero-ADP	
	WP	ADP	WP	ADP	WP	ADP	WP	ADP
AT-Dec-POSG	—	—	0.5152	0.3183	0.5005	0.1813	0.5472	0.0143
PerfectDou	0.4848	-0.3183	—	—	0.5430	0.1430	0.5120	0.0610
DouZero-WP	0.4995	-0.1813	0.4570	-0.1430	—	—	0.5210	-0.0380
DouZero-ADP	0.4528	-0.0143	0.4880	-0.0610	0.4790	0.0380	—	—

Table 2: Doudizhu ablation against the fixed Base DMC learner. Entries report mean $\pm$ std over three matched seeds.

Variant	WP (%) vs Base	ADP vs Base	Gain vs Base
Base + Module A	56.00 $\pm$ 1.50	0.46 $\pm$ 0.03	+3.90 WP / +0.15 ADP
Base + Module B	58.61 $\pm$ 1.12	0.66 $\pm$ 0.07	+6.51 WP / +0.35 ADP
Base + Module A + Module B	58.90 $\pm$ 1.40	0.60 $\pm$ 0.04	+6.80 WP / +0.29 ADP
Full	59.70 $\pm$ 1.10	0.67 $\pm$ 0.05	+7.60 WP / +0.36 ADP

Appendix A.13 gives a finite-space proof. This is stronger than a mere “public actions can carry some bits” statement: after fixing the evaluation occupancy and an action-relevant summary, if legal teammate-side information is too weak to identify the summary needed by the centralized continuation, some part of $\varepsilon_{\text{coord}}^{\pi}$ is unavoidable even if Module B improves C_{legal} . The fixed-adversary-pool league-monotonicity guarantee used to motivate Module C is deferred to Appendix A.9.

Appendix A.15 and Appendix A.16 record two additional supporting results: an occupancy-average public-channel action-information statement and the residual-based approximate-equilibrium decomposition. We treat them as bookkeeping tools rather than main-text centerpieces; the main structural claims here are that C4 induces a public-belief control problem, richer legal public signals are weakly better than coarser ones, and coarse legal information implies irreducible coordination residue. Proposition 9 in the appendix also records an optional recovery-rate bridge linking legal signaling budget to ρ_{coord} under stronger calibration assumptions.

6 Experiments

This section tests one prediction: Doudizhu and the replay-based multi-contract execution stress test should exhibit the same public-action coordination bottleneck under AT-Dec-POSG.

Setup. The two domains intentionally use different base learners because their action spaces differ: Doudizhu uses a DMC-style per-position Q-network with a separate bidding network, while the replay-based multi-contract execution stress test uses a PPO-style actor-critic. Because the underlying finance-side market data and held-asset details are proprietary and cannot be directly released, the second domain is instantiated as a replay-based simulated environment rather than as an openly disclosed live trading task. Supporting protocol, training-curve, and action-level details are deferred to Appendix A.20, Appendix A.19, Appendix A.18, and Appendix A.28.

Benchmarks and baselines. The Doudizhu study is the primary benchmark and uses reproduced public opponents under one bidding-enabled, seat-averaged 10k-deck evaluation script. We test whether the full system beats strong baselines, whether the Base / +A / +B / +A+B / Full ablation follows the intended A/B/C split, and whether the replay-based execution stress test reproduces the same public-channel recovery pattern; compute and inherited-default details are deferred to Appendix A.24.

Evaluation scripts. Table 1 uses a seat-averaged bidding-enabled 10k-deck tournament against reproduced public opponents; Table 2 reports internal ablations against a fixed Base DMC learner over three matched seeds rather than a round-robin; Table 3 uses the matched replay benchmark over three matched seeds; and Tables 4–5 use matched C4-compliance/intervention scripts plus a held-out probe protocol. Appendix A.28 gives the full protocol matrix.

Main results. Here Base denotes the underlying DMC learner and serves as the fixed opponent for Table 2. The stand-alone Base + Module B row isolates legal public-trace coordination. For reference,

Table 3: Replay-based multi-contract execution results (mean $\pm$ std over three matched seeds; lower Cost, Shortfall, and Trace-frag. are better, with low Trace-frag. indicating greater robustness to public-trace corruption).

Method	Cost	Shortfall	RA-PnL	Comp. rate	Trace-frag.
PPO	6.399 $\pm$ 0.448	5.754 $\pm$ 0.403	-5.813 $\pm$ 0.407	0.952 $\pm$ 0.067	0.380 $\pm$ 0.003
MAPPO	5.823 $\pm$ 0.225	6.444 $\pm$ 0.331	-4.585 $\pm$ 0.221	0.965 $\pm$ 0.071	0.347 $\pm$ 0.012
Full	3.895 $\pm$ 0.273	3.291 $\pm$ 0.230	-3.356 $\pm$ 0.235	0.977 $\pm$ 0.068	0.256 $\pm$ 0.046
Full incumbent	3.931 $\pm$ 0.275	3.163 $\pm$ 0.221	- 3.246 $\pm$ 0.227	0.957 $\pm$ 0.067	0.255 $\pm$ 0.046

Table 4: Cross-domain coordination summary for the C4-compliant A+B stack.

Domain	No coord.	Legal coord. (A+B)	Explicit latent	Δ_{unrec}	ρ_{coord}	Interv. gap (A+B)	Interv. gap (ctrl)
Doudizhu	0.521	0.589	0.612	0.023	74.7%	0.19	0.04
Multi-contract execution	0.560	0.760	0.840	0.080	71.4%	0.17	0.03

Base + Module C reaches WP 59.30% and ADP 0.63, but this should be read as a heavier-training robustness effect rather than as matched-compute legal coordination recovery. For execution, we report the completion-qualified lowest-Cost checkpoint whose completion matches or exceeds PPO, keep the earlier incumbent as a nearby Pareto comparison, and defer the full sweep to Appendix A.30.

Against reproduced public opponents, the full Doudizhu system reaches WP 0.5152 (95% paired-bootstrap CI [0.5078, 0.5226]) and ADP 0.3183 against PerfectDou, fully surpassing the current SOTA PerfectDou baseline under the shared protocol. In Table 3, lower Cost, Shortfall, and Trace-frag. are better, with Trace-frag. measuring robustness to public-trace corruption. PPO and MAPPO [25] define weaker trade-offs, while Full is strongest overall. Full is the completion-qualified lowest-Cost point; Full incumbent is a nearby Pareto variant with better shortfall and risk-adjusted PnL but lower completion, with the longer sweep deferred to Appendix A.30. We do not compare raw WP and shortfall directly: Δ_{coord} denotes the total explicit-minus-no-coordination gap, Δ_{unrec} the explicit-minus-legal unrecovered residue, and $\rho_{\text{coord}} = 1 - \Delta_{\text{unrec}}/\Delta_{\text{coord}}$. Here Perf is used only for the C4-compliance protocol: seat-averaged win rate in Doudizhu and normalized slippage reduction in replay-based execution. In Table 4, the legal row corresponds to the C4-compliant Base + Module A + Module B stack in both domains, so ρ_{coord} should be read relative to the chosen explicit-latent upper-bound architecture rather than as an absolute domain constant. Full metric definitions appear in Appendix A.29.

Cross-domain coordination pattern. Table 4 is the compact cross-domain comparison for the matched C4-compliant A+B stack. Across both domains, the C4-compliant A+B variant recovers a nonzero fraction of the explicit-latent coordination benefit, and teammate-trace corruption hurts it far more than the matched-capacity control. Table 3 reports the deployed checkpoint’s public-trace corruption fragility as a robustness score, so lower Trace-frag. means more robust execution under public-trace corruption; Table 4 instead uses the A+B-versus-control intervention-gap contrast as mechanism evidence, so a larger gap means stronger reliance on the legal public trace.

Legal-capacity probe. Table 5 reports a held-out lower bound on how much teammate-relevant sender-private structure becomes decodable from the legal public trace. Fix a coordination-critical slice C_t , a sender-private summary Z_t , and receiver-local information x_t^j . We define $C_{\text{legal}}^{\text{lb}} := I(Z_t; h_t^{\text{pub}} | x_t^j, C_t)$ and report the proxy $C_{\text{legal}}^{\text{lb}} := H(Z_t | x_t^j, C_t) - \mathcal{L}_{\text{rec}}^{\text{held}}$, where $\mathcal{L}_{\text{rec}}^{\text{held}}$ is the held-out cross-entropy of receiver probe $q_{\psi}(z | x_t^j, h_t^{\text{pub}}, C_t)$. The no-public control isolates the share attributable to h_t^{pub} . In Doudizhu, Z_t is the Module-B hand-strength / remaining-hand target; in multi-contract execution, it is the sender urgency bin.

Table 5: Held-out legal-capacity probe on coordination-critical histories.

Domain	Variant	Receiver CE $\downarrow$	Decode Acc. $\uparrow$	$C_{\text{legal}}^{\text{lb}} \uparrow$
Doudizhu	Base	0.931	0.574	0.061
Doudizhu	Base + A + B	0.642	0.731	0.350
Doudizhu	Full	0.631	0.737	0.361
Multi-contract execution	Base	0.884	0.591	0.048
Multi-contract execution	Base + A + B	0.603	0.748	0.329
Multi-contract execution	Full	0.592	0.753	0.340

383 **Scope and structural reading.** Most legal-capacity gain is already realized by A+B, so Table 4
384 is the main cross-domain legal-coordination evidence chain and Table 5 shows that most decod-
385 able public-channel gain appears before adding Module C; residual bookkeeping and extended
386 Pareto/support analyses are deferred to Appendix A.16, Appendix A.30, and Appendix A.31.

387 7 Conclusion

388 We identify an overlooked but practically important C4 information constraint in adversarial team
389 games: teammates may share reward yet be unable to use any private execution-time channel. We then
390 give a C4-compliant legal public-trace coordination framework. The structural theory explains why,
391 once private teammate channels are disallowed, the public trace is the only remaining coordination
392 medium, and the experiments show the same coordination-recovery pattern in Doudizhu and a
393 replay-based execution stress test. On Doudizhu, the full system reaches WP 0.5152 and ADP 0.3183
394 against PerfectDou, beating the current SOTA baseline under the shared bidding-enabled protocol.
395 Across Doudizhu and the replay-based execution stress test, the C4-compliant A+B variant recovers
396 74.7% and 71.4% of the explicit-latent coordination benefit, while intervention and legal-capacity
397 diagnostics support the same mechanism reading, and Full mainly adds league-side robustness.
398 This legality-first perspective may help constrained coordination settings, but it can also strengthen
399 competitive automation in market-like environments and therefore warrants cautious deployment.

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522 A Additional Proofs and Experimental Notes

523 A.1 Proof of Lemma 1

524 By definition, $\Pi_{\mathcal{M}}^{\text{Dec}} \subseteq \Pi_{\mathcal{M}}$. Therefore the maximization over $\Pi_{\mathcal{M}}^{\text{Dec}}$ is taken over a subset of the
 525 feasible centralized team policies, which immediately yields

$$\sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}} \inf_{\pi_{\mathcal{K}}} V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}(s) \leq \sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}} \inf_{\pi_{\mathcal{K}}} V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}(s).$$

526 When the centralized optimum is attained, equality requires that an optimal centralized team policy
 527 admit a decentralized realization using no teammate-private information beyond the public history.

528 A.2 A strict-gap construction for Lemma 1

529 We give a minimal one-step AT-Dec-POSG in which the inequality of Lemma 1 is strict at a fixed
 530 root state.

531 Let the team consist of two teammates $i \in \{1, 2\}$ and let the adversarial side contain a dummy
 532 opponent with no nontrivial action. The episode starts from a fixed root state s_0 . From s_0 , a public
 533 chance transition draws a latent pair $s = (b_1, b_2) \in \{0, 1\}^2$ uniformly, where teammate i observes
 534 only its own bit b_i . There is no informative public history before the decision. Each teammate chooses
 535 an action $a_i \in \{0, 1\}$. The team reward is

$$r(s, a_1, a_2) = \mathbf{1}[a_1 = a_2 = b_1 \oplus b_2],$$

536 and the adversary receives the negative reward.

537 An unrestricted team policy that observes both bits can achieve value 1 by setting $a_1 = a_2 = b_1 \oplus b_2$.
 538 Under C4, the one-step expected payoff is multilinear in the teammates' product action kernels, so an
 539 optimum over decentralized policies is attained at a deterministic extreme point. It therefore suffices
 540 to analyze deterministic response functions $a_1 = f_1(b_1)$ and $a_2 = f_2(b_2)$. To obtain reward 1 on all
 541 four states, we would need

$$f_1(0) = f_2(0) = 0, \quad f_1(0) = f_2(1) = 1, \quad f_1(1) = f_2(0) = 1, \quad f_1(1) = f_2(1) = 0,$$

542 which is impossible because the first two equalities already force $f_1(0)$ to be both 0 and 1. Hence
 543 no C4-compliant decentralized policy can succeed on all four states; in fact, at most two of the four
 544 constraints can be satisfied simultaneously, so the best decentralized value from s_0 is at most $1/2$.
 545 Therefore

$$\sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}} \inf_{\pi_{\mathcal{K}}} V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}(s_0) < \sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}} \inf_{\pi_{\mathcal{K}}} V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}(s_0),$$

546 which witnesses a strict coordination gap induced purely by C4.

547 A.3 A conservative upper envelope for C4-induced loss

548 We now record a complementary upper-side statement. Unlike Lemma 1, which compares minimax
 549 values directly, the bound below is stated first against a fixed adversary policy. This is the setting
 550 most closely matched by the experimental coordination-gap protocol, where explicit-latent, legal, and
 551 no-coordination variants are evaluated against the same opponent pool.

552 **Definition 2** (Fixed-adversary coordination loss). *For a fixed adversary policy $\pi_{\mathcal{K}}$, define*

$$\Delta_{\text{C4}}(s; \pi_{\mathcal{K}}) = \sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}} V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}(s) - \sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}} V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}(s).$$

553 **Proposition 4** (Conservative upper envelope via coordination-critical occupancy). *Assume a finite*
 554 *horizon H and per-step team rewards $r_t \in [R_{\min}, R_{\max}]$, and write*

$$R_{\text{span}} = R_{\max} - R_{\min}.$$

555 *Fix a centralized team benchmark $\pi_{\mathcal{M}}^{\text{cen}}$ and an adversary policy $\pi_{\mathcal{K}}$. Let $\mathcal{C}$ be a measurable set of*
 556 *team decision histories with the following property: there exists a decentralized policy $\tilde{\pi}_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}$*
 557 *that agrees with $\pi_{\mathcal{M}}^{\text{cen}}$ on every team decision history outside $\mathcal{C}$. Define the coordination-critical hit*
 558 *event*

$$E_{\mathcal{C}} = \{\text{the trajectory visits } \mathcal{C}\}, \quad q_{\text{crit}} = \mathbb{P}_{\pi_{\mathcal{M}}^{\text{cen}}, \pi_{\mathcal{K}}}(E_{\mathcal{C}} \mid s).$$

559 Then

$$V^{\pi_{\mathcal{M}}^{\text{cen}}, \pi_{\mathcal{K}}}(s) - V^{\tilde{\pi}_{\mathcal{M}}, \pi_{\mathcal{K}}}(s) \leq HR_{\text{span}} q_{\text{crit}}.$$

560 Consequently, if $\pi_{\mathcal{M}}^{*, \pi_{\mathcal{K}}} \in \arg \max_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}} V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}(s)$, then

$$\Delta_{\text{C4}}(s; \pi_{\mathcal{K}}) \leq HR_{\text{span}} q_{\text{crit}}$$

561 for any decentralized realization $\tilde{\pi}_{\mathcal{M}}$ of $\pi_{\mathcal{M}}^{*, \pi_{\mathcal{K}}}$ that can mismatch only on $\mathcal{C}$.

562 **Proof.** Couple the trajectories induced by $(\pi_{\mathcal{M}}^{\text{cen}}, \pi_{\mathcal{K}})$ and $(\tilde{\pi}_{\mathcal{M}}, \pi_{\mathcal{K}})$ with the same exogenous
 563 randomness. Because the two team policies agree on every team decision history outside $\mathcal{C}$, the
 564 coupled trajectories are identical up to the first visit to $\mathcal{C}$. Hence, on the complement event $E_{\mathcal{C}}^c$, the
 565 two finite-horizon returns are exactly the same.

566 On the event $E_{\mathcal{C}}$, the two trajectories may diverge after the first visit to $\mathcal{C}$. Regardless of what happens
 567 afterward, the difference between two finite-horizon returns is at most the total reward range

$$HR_{\text{span}}.$$

568 Therefore

$$V^{\pi_{\mathcal{M}}^{\text{cen}}, \pi_{\mathcal{K}}}(s) - V^{\tilde{\pi}_{\mathcal{M}}, \pi_{\mathcal{K}}}(s) \leq HR_{\text{span}} \mathbb{P}_{\pi_{\mathcal{M}}^{\text{cen}}, \pi_{\mathcal{K}}}(E_{\mathcal{C}} \mid s) = HR_{\text{span}} q_{\text{crit}}.$$

569 For the second claim, let $\pi_{\mathcal{M}}^{*, \pi_{\mathcal{K}}}$ be a centralized maximizer against $\pi_{\mathcal{K}}$. Since

$$\sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}} V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}(s) \geq V^{\tilde{\pi}_{\mathcal{M}}, \pi_{\mathcal{K}}}(s),$$

570 we obtain

$$\Delta_{\text{C4}}(s; \pi_{\mathcal{K}}) \leq V^{\pi_{\mathcal{M}}^{*, \pi_{\mathcal{K}}}, \pi_{\mathcal{K}}}(s) - V^{\tilde{\pi}_{\mathcal{M}}, \pi_{\mathcal{K}}}(s) \leq HR_{\text{span}} q_{\text{crit}}.$$

571

□

572 **Interpretation.** The proposition is intentionally conservative. The set $\mathcal{C}$ may be any measurable
 573 superset of the truly non-decentralizable histories of the centralized benchmark, so the bound remains
 574 valid even when $\mathcal{C}$ is only a coarse approximation of the public-history ambiguity set. Its message is
 575 that large coordination loss requires large occupancy of histories at which public history and local
 576 observations are insufficient to reproduce the centralized prescription.

577 A.4 Full framework background and inherited conditions

578 The main text compresses the formal background to fit the nine-page storyline. This appendix
 579 subsection records the longer framework details.

580 **Relation to POSGs and ATMGs.** AT-Dec-POSG is a constrained subclass of POSGs and adversar-
 581 ial team Markov games, introduced not for additional expressivity but for analytical specificity. The
 582 key point is that the conjunction of a team–adversary partition, partial observability (C3), and strict
 583 intra-team execution-time isolation (C4) changes the admissible team policy class in an operationally
 584 nontrivial way: the team may share reward but may not share private state, so coordination must
 585 be mediated entirely through the public interaction history. This creates a distinct learning problem
 586 relative to standard CTDE or explicit-communication formulations, because the relevant gap is not
 587 whether coordination is useful in principle, but how much value can be recovered within the legally
 588 admissible decentralized class.

Table 6: Relationship between nearby frameworks. AT-Dec-POSG narrows the ambient class by making the team-versus-adversary partition, partial observability, and strict execution-time information isolation simultaneously explicit.

Framework	Team-vs-adversary	Partial observability	Explicit C4-style isolation
POSG	optional	yes	no
ATMG	yes	not central	no
AT-Dec-POSG	yes	yes	yes

589 **Reading Table 6.** This table is included only to make the ambient class relationship explicit. Its
 590 role is not to claim that AT-Dec-POSG is more general than POSGs or ATMGs, but to show that
 591 the paper isolates the specific conjunction of team–adversary structure, partial observability, and
 592 execution-time teammate isolation that drives the later Module A/B/C design.

593 **Stochastic game baseline.**

594 **Definition 3** (Stochastic game). *A stochastic game is a tuple*

$$\mathcal{G} = \langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}_{i \in \mathcal{N}}, \mathcal{T}, \{r_i\}_{i \in \mathcal{N}}, \gamma \rangle,$$

595 where $\mathcal{N}$ is the set of agents, $\mathcal{S}$ is the latent state space, $\mathcal{A}_i$ is agent i 's action space, $\mathcal{T}$ is the
 596 transition kernel, r_i is the reward of agent i , and $\gamma \in [0, 1)$ is the discount factor.

597 **Adversarial team Markov games.** An adversarial team Markov game augments the stochastic
 598 game with a team partition $\mathcal{N} = \mathcal{M} \cup \mathcal{K}$ and a zero-sum team-versus-adversary reward structure.
 599 In the AT-Dec-POSG specialization below, the team side is summarized by one scalar reward r_t , so
 600 teammates satisfy $r_i = r_t$ for $i \in \mathcal{M}$, while the adversarial side is evaluated in the opposing payoff
 601 coordinates $-r_t$. In the present paper, we treat this ATMG backbone as contributing two inherited
 602 structural conditions:

- 603 1. **C1 (team-versus-adversary partition):** there exists a distinguished team $\mathcal{M}$ and an opposing
 604 side $\mathcal{K}$.
- 605 2. **C2 (shared team objective):** agents in $\mathcal{M}$ optimize a shared team objective, and that team
 606 objective is opposed to the adversarial side in the team-versus-adversary sense.

607 Existing formulations of this kind are the closest conceptual predecessor of our setting, but typically
 608 assume perfect information or do not make teammate-private execution-time information isolation
 609 explicit.

610 **Full AT-Dec-POSG tuple.**

611 **Definition 4** (AT-Dec-POSG). *An Adversarial Team Decentralized Partially Observable Stochastic*
 612 *Game is a tuple*

$$\mathcal{G}_{\text{AT-Dec}} = \langle \mathcal{M}, \mathcal{K}, \mathcal{S}, \{\mathcal{O}_i\}_{i \in \mathcal{N}}, \{\Omega_i\}_{i \in \mathcal{N}}, \{\mathcal{A}_i\}_{i \in \mathcal{N}}, \mathcal{T}, r, \gamma \rangle$$

613 with two additional constraints:

- 614 1. **C3 (partial observability):** $o_t^i = \Omega_i(s_t) \in \mathcal{O}_i$, where the observation map is non-identity and in
 615 general insufficient to reconstruct the full latent state.
- 616 2. **C4 (strict intra-team isolation):** for distinct teammates $i, j \in \mathcal{M}$, policy π_i may condition only
 617 on $\tau_t^i = (h_t^{\text{pub}}, x_t^i)$, never on teammate-private observation o_t^j or on any private execution-time
 618 message not already present in the environment.

619 Relative to adversarial team games satisfying C1–C2, AT-Dec-POSG introduces two additional
 620 information-structure restrictions: observation is no longer equal to state (C3), and teammate-private
 621 information cannot be shared at execution time (C4). Together these conditions restrict the feasible
 622 team policy class to policies that coordinate only through public interaction.

623 A.5 Doudizhu instantiation details

624 **Restatement of Example 1.** Doudizhu is best treated as a public role-assignment AT-Dec-POSG
 625 continuation game rather than as a fixed farmer-versus-landlord partition from time zero. The public
 626 bidding stage first generates a terminal bid history h^{bid} ; after that public pre-play stage, a realized
 627 role assignment is fixed and the continuation game becomes the AT-Dec-POSG proper.

628 **Public role-assignment stage.** Let $\mathcal{N}$ be the ex ante player set and let $\mathcal{R}$ be a finite set of role
 629 assignments. A public role-assignment kernel

$$\kappa(r \mid h^{\text{bid}})$$

maps the terminal public bidding history h^{bid} to a realized role assignment $r \in \mathcal{R}$. For each realized r , define the team and adversarial sides as $\mathcal{M}(r) \subset \mathcal{N}$ and $\mathcal{K}(r) = \mathcal{N} \setminus \mathcal{M}(r)$. The AT-Dec-POSG proper is the continuation game conditioned on (r, h^{bid}) , with initial public history

$$h_0^{\text{pub}} = (h^{\text{bid}}, r).$$

For $t \geq 0$, teammate $i \in \mathcal{M}(r)$ may condition only on $\tau_t^i = (h_t^{\text{pub}}, x_t^i)$, never on teammate-private state or on a private execution-time message.

Doudizhu-specific realization. In Doudizhu, $\mathcal{N}$ is the three-seat player set and $\mathcal{R}$ is the finite set of landlord assignments. The public bidding stage generates h^{bid} , and the role-assignment kernel is degenerate at the realized landlord role L . The continuation AT-Dec-POSG is then defined with

$$\mathcal{K}(L) = \{L\}, \quad \mathcal{M}(L) = \mathcal{N} \setminus \{L\},$$

farmer-side local observation equal to private hand plus inherited public bid/play history, and shared farmer reward given by the win/loss outcome from the farmer perspective.

Why bidding is treated as legal public pre-history. Before bidding ends there is no fixed farmer team, so C1, C2, and C4 should not be read in farmer-versus-landlord form throughout the bidding stage. Instead, bidding is a public role-assignment stage whose realized output (L, h^{bid}) initializes the continuation game. The entire bid sequence is then inherited as legal public history during play. This is exactly why bids can carry information into the continuation without violating C4: the signal is public before and after role assignment, and no teammate-private state or private message is introduced.

Conditioning of the structural results. All AT-Dec-POSG structural results in the paper are applied to the continuation game after public role assignment. Equivalently, Lemma 1 and the later structural propositions are interpreted conditionally on the realized role assignment and inherited public bid history (L, h^{bid}) . This resolves the apparent team-partition issue cleanly: the paper does not assume that the farmer-versus-landlord partition is already fixed during bidding, only that the continuation game after public role assignment satisfies the AT-Dec-POSG definition.

Module-B legality under inherited public history. Module B remains legal in this role-assignment view because its deployed actors use only the public bidding/play trace, not teammate-private bidding features or a private execution-time channel. In implementation, bidding uses a separate network because the action alphabet is different, but both the bid trace and the play trace enter only as public history inherited by the continuation farmers.

Proof sketch. Conditioned on a realized landlord role L and public bid history h^{bid} , Doudizhu satisfies the definition by taking the latent state to include all remaining hands and the evolving public history, taking each realized farmer’s local observation to be the projection of that state onto its private hand and the inherited public bid/play trace, and defining the team reward from the farmers’ shared perspective. The landlord is evaluated in the opposing payoff coordinate. C4 holds in the continuation game because one farmer never observes the other farmer’s private hand, and the only information inherited from bidding is the public history already contained in h_0^{pub} .

Doudizhu reward specification used in code. Let $w \in \{\text{landlord}, \text{farmer}\}$ denote the winner side and let $i \in \{\text{landlord}, \text{landlord_up}, \text{landlord_down}\}$ denote a seat. The play-phase reward is seat-coordinate aligned through

$$\mathbf{1}_{\text{win}}(i) = \mathbf{1}[(w = \text{landlord}) = (i = \text{landlord})].$$

For objective ‘wp’, the terminal play reward is

$$r_{\text{play}}^i = \begin{cases} 1 - 0.0033 \text{ step_count}, & \mathbf{1}_{\text{win}}(i) = 1, \\ -(1 - 0.0033 \text{ step_count}), & \mathbf{1}_{\text{win}}(i) = 0. \end{cases}$$

For ‘adp’ and ‘logadp’, the same winner/loser sign convention is used, multiplied by the objective-specific bomb and multiply factors implemented in the environment. Therefore both farmer seats receive positive reward when farmers win and negative reward when landlord wins.

672 The bidding head uses a landlord-coordinate scalar

$$r_{\text{bid}}^L \in \left\{ 2^{(\text{bid_count}-1)/8}, -2^{(\text{bid_count}-1)/8} \right\},$$

673 and learner targets convert non-landlord bid types by sign so that bidding and play remain consistent
674 in side semantics.

675 A.6 Additional cross-domain instantiation details

676 This subsection records the longer cross-domain instantiation discussion that supports the abbreviated
677 main-text summary in Section 3.

678 **Replay-based structural stress test.** We instantiate the second domain as a replay-based structural
679 stress test over a fixed universe of synchronized futures contracts. Each execution agent controls one
680 contract or one portfolio slice, observes only local order-book state, local inventory, remaining target
681 quantity, and time-to-go, and chooses among a discrete family of passive, neutral, and aggressive
682 execution templates with associated size buckets. Public observations consist of market-visible orders,
683 trades, and cancellations across the contract set; teammate-private intentions, hidden target schedules,
684 and internal latent states are never exposed on the execution path. The team reward is the negative
685 aggregate implementation shortfall together with inventory and non-completion penalties, while
686 environment-side pressure is represented by the aggregate liquidity response implicit in replayed
687 order flow: fill rates, price impacts, and queue dynamics depend on how aggressively the execution
688 team participates. Operationally, the central coordination question is whether one execution agent
689 can reveal its current execution cadence or urgency to the rest of the team through these limited
690 public actions. This gives a concrete finance-side stress test of C3 and C4 with a fixed replay engine,
691 preprocessing pipeline, and train/validation/test split.

692 **Formal instantiation claim under a task-level adversary abstraction.** Under a task-level ad-
693 versary abstraction, the replay-based multi-contract execution benchmark admits an AT-Dec-POSG
694 instantiation. Concretely, the team is

$$\mathcal{M} = \{\text{exec}_1, \dots, \text{exec}_n\},$$

695 the task-level adversarial side is represented by the aggregate liquidity response

$$\mathcal{K} = \{\text{market_side}\},$$

696 each execution agent i observes only its contract-local order book, local inventory state, remaining
697 target, time-to-go, and the public market tape, and the shared team reward is the negative aggregate
698 execution shortfall together with inventory-risk and non-completion penalties. C3 holds because no
699 single execution agent observes the full multi-contract latent state. C4 holds because exec_i cannot
700 access the private latent state of exec_j , such as pending orders, internal signals, or inventory targets;
701 the only legal interaction channel is the public market tape.

702 **On the abstraction.** This claim is intentionally narrower than the Doudizhu instantiation. Unlike
703 Doudizhu, the finance-side opponent is not a single explicitly parameterized player acting in the same
704 online training loop. Instead, the market-liquidity side is aggregated into one task-level adversary
705 whose pressure is expressed through replayed order placement, fills, and queue movement. This
706 abstraction is enough for the information-structure claim needed in the paper, but it should not be
707 read as evidence that the second domain is a perfectly symmetric learning-adversary counterpart to
708 Doudizhu.

709 **Cross-domain symmetry of the coordination problem.** The domain semantics differ, but the
710 public-action coordination bottleneck is the same. In Doudizhu, the sender uses finite public actions
711 to indicate whether it holds high-card control or other private hand strength relevant to the joint line of
712 play. In multi-contract execution, the sender uses finite public actions to indicate its current execution
713 cadence or urgency. In both cases, the team must transmit private-state information through a limited
714 public action channel while operating under external pressure, which is why a single C4-compliant
715 coordination mechanism and a single public-channel proposition can cover both domains at the level
716 of information structure.

Table 7: Structural mapping from the two application domains to AT-Dec-POSG.

Component	Doudizhu	Multi-contract futures execution
Team $\mathcal{M}$	Realized farmers after bidding	Multiple execution agents
Adversary $\mathcal{K}$	Realized landlord after bidding	Task-level market-side liquidity abstraction
Local observation	Own hand + inherited public bid/play history	Local order book + local inventory + public tape
Public channel	Bids, played cards, passes	Orders, trades, cancellations
Team reward	Shared win/loss payoff	Shared shortfall plus risk penalty
C4 restriction	No access to teammate's private hand	No access to teammate's private execution state

A.7 Domain-mapping table

Reading Table 7. The purpose of this mapping is to show that the two application domains differ in action alphabet and base learner while still instantiating the same information-structure bottleneck. In both rows, teammate-private state is hidden at execution time and only public actions remain as admissible coordination carriers, which is the shared structural premise used by the main text.

A.8 Execution-time legality of Module B

Proposition 5 (C4 legality of public-trace coordination). *Assume the execution policy of each teammate is implemented as $a_t^i \sim \pi_i(\cdot \mid \tau_t^i)$, where π_i may be stochastic or degenerate at a greedy action and $\tau_t^i = (h_t^{\text{pub}}, x_t^i)$ is the teammate's full legal history. Assume also that all sender/receiver auxiliary logits, targets, latents, and privileged features are used only through training losses and are absent from the deployed actor input. Then Module B does not violate C4: for each teammate j , the execution-time action a_t^j is measurable with respect to $\sigma(\tau_t^j)$ and depends on the rest of the team only through the legal public history.*

By assumption, teammate j 's execution policy is the stochastic kernel $\pi_j(\cdot \mid \tau_t^j)$. Hence the sampled action a_t^j is measurable with respect to $\sigma(\tau_t^j)$. Any sender/receiver auxiliary logit, target, latent, teacher feature, or privileged tensor used during training is absent from the execution path, so dependence on teammate i can enter only through the public history already contained in τ_t^j . No teammate-private observation or private execution-time message is added to the deployed actor input. Therefore Module B satisfies C4 at execution time.

A.9 Restricted-game improvement under league expansion

Proposition 6 (Restricted-league monotonicity). *Let $\Pi_{\mathcal{M}}^L$ and $\Pi_{\mathcal{K}}^L$ be the current finite policy pools for the team and the adversary, and define the restricted-game value*

$$v^L = \max_{\mu \in \Delta(\Pi_{\mathcal{M}}^L)} \min_{\nu \in \Delta(\Pi_{\mathcal{K}}^L)} \mathbb{E}_{\pi_{\mathcal{M}} \sim \mu, \pi_{\mathcal{K}} \sim \nu} [V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}] .$$

If the league is expanded by adding any new team policy $\pi_{\mathcal{M}}^{\text{new}}$ to the pool, so that $\Pi_{\mathcal{M}}^{L+1} = \Pi_{\mathcal{M}}^L \cup \{\pi_{\mathcal{M}}^{\text{new}}\}$, then

$$v^{L+1} \geq v^L .$$

Define the restricted-game payoff

$$f^L(\mu, \nu) := \mathbb{E}_{\pi_{\mathcal{M}} \sim \mu, \pi_{\mathcal{K}} \sim \nu} [V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}] .$$

Because $\Pi_{\mathcal{M}}^L$ and $\Pi_{\mathcal{K}}^L$ are finite, the mixed-strategy sets $\Delta(\Pi_{\mathcal{M}}^L)$ and $\Delta(\Pi_{\mathcal{K}}^L)$ are compact convex simplices, and f^L is continuous and bilinear in (μ, ν) . Therefore the finite zero-sum restricted game satisfies the minimax theorem:

$$v^L = \max_{\mu \in \Delta(\Pi_{\mathcal{M}}^L)} \min_{\nu \in \Delta(\Pi_{\mathcal{K}}^L)} f^L(\mu, \nu) = \min_{\nu \in \Delta(\Pi_{\mathcal{K}}^L)} \max_{\mu \in \Delta(\Pi_{\mathcal{M}}^L)} f^L(\mu, \nu) .$$

Now expand the team pool to

$$\Pi_{\mathcal{M}}^{L+1} = \Pi_{\mathcal{M}}^L \cup \{\pi_{\mathcal{M}}^{\text{new}}\} ,$$

746 while holding the adversary pool fixed. Let

$$f^{L+1}(\mu, \nu) := \mathbb{E}_{\pi_{\mathcal{M}} \sim \mu, \pi_{\mathcal{K}} \sim \nu} [V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}]$$

747 denote the corresponding payoff on $\Delta(\Pi_{\mathcal{M}}^{L+1}) \times \Delta(\Pi_{\mathcal{K}}^L)$. Every old team mixture remains feasible
 748 after expansion by assigning zero mass to $\pi_{\mathcal{M}}^{\text{new}}$, so $\Delta(\Pi_{\mathcal{M}}^L)$ embeds as a face of $\Delta(\Pi_{\mathcal{M}}^{L+1})$. Hence,
 749 for every fixed adversary mixture $\nu \in \Delta(\Pi_{\mathcal{K}}^L)$,

$$\max_{\mu \in \Delta(\Pi_{\mathcal{M}}^{L+1})} f^{L+1}(\mu, \nu) \geq \max_{\mu \in \Delta(\Pi_{\mathcal{M}}^L)} f^L(\mu, \nu),$$

750 because the maximization on the left is taken over a superset of feasible team mixtures and agrees
 751 with f^L on the embedded old face. Taking the minimum over the unchanged adversary simplex yields

$$\min_{\nu \in \Delta(\Pi_{\mathcal{K}}^L)} \max_{\mu \in \Delta(\Pi_{\mathcal{M}}^{L+1})} f^{L+1}(\mu, \nu) \geq \min_{\nu \in \Delta(\Pi_{\mathcal{K}}^L)} \max_{\mu \in \Delta(\Pi_{\mathcal{M}}^L)} f^L(\mu, \nu).$$

752 Applying the minimax theorem to both restricted games gives

$$v^{L+1} \geq v^L.$$

753 Thus, when the adversary pool is held fixed, enlarging the team-side policy pool cannot decrease the
 754 restricted minimax value. $\square$

755 A.10 Supporting lemma: non-positive shaping bonus

756 If $V_{\text{farmer-joint}}(s) \in [0, 1]$ and $\max_a Q_L(o, a) \in [0, 1]$, then for $\alpha \geq 0$ and $\gamma \in [0, 1)$, the products
 757 $\alpha\gamma V_{\text{farmer-joint}}(s)$ and $\alpha\gamma \max_a Q_L(o, a)$ are non-negative. Negating them yields non-positive bonus
 758 terms.

759 A.11 Proof of Proposition 1

760 **Finite-space convention for this subsection.** Assume a finite horizon H , finite state/action al-
 761 phabets, and finite teammate-private alphabets. Fix an adversary policy $\pi_{\mathcal{K}}(a_t^{\mathcal{K}} \mid s_t, h_t^{\text{pub}})$ with
 762 no additional hidden memory beyond the current latent/public state. Once this policy is fixed, the
 763 team best-response problem becomes an ordinary finite-horizon stochastic control problem with
 764 decentralized information constraints. All posteriors, prescriptions, and conditional laws below are
 765 therefore well defined on finite sets.

766 **Common and private information split.** The statement is a direct specialization of the common-
 767 information dynamic-programming reduction for decentralized control with partial history sharing to
 768 the AT-Dec-POSG legality constraint C4. The team best-response problem has

- 769 • common information equal to the public history h_t^{pub} ,
- 770 • private information equal to the teammate-local components x_t^i ,
- 771 • decentralized actions restricted to measurable functions of $(h_t^{\text{pub}}, x_t^i)$.

772 Define the coordinator state as the posterior

$$b_t = \mathbb{P}(s_t, x_t^1, \dots, x_t^{|\mathcal{M}|} \mid h_t^{\text{pub}}).$$

773 **Coordinator reformulation.** At each stage t , define a team-side prescription as a tuple

$$\Gamma_t = (\gamma_t^i)_{i \in \mathcal{M}}, \quad \gamma_t^i : \mathcal{X}_t^i \rightarrow \Delta(\mathcal{A}_t^i),$$

774 where $\mathcal{X}_t^i$ is the finite private-information alphabet of teammate i . Given b_t and Γ_t , the induced
 775 decentralized action law is

$$\mathbb{P}(a_t \mid b_t, \Gamma_t) = \sum_{s_t, x_t^1, \dots, x_t^{|\mathcal{M}|}} b_t(s_t, x_t^1, \dots, x_t^{|\mathcal{M}|}) \prod_{i \in \mathcal{M}} \gamma_t^i(a_t^i \mid x_t^i).$$

776 Because the game model and $\pi_{\mathcal{K}}$ are common knowledge and $\pi_{\mathcal{K}}$ carries no hidden memory beyond
 777 (s_t, h_t^{pub}) , Bayes' rule yields a closed update

$$b_{t+1} = F_t(b_t, \Gamma_t, h_{t+1}^{\text{pub}})$$

778 for some deterministic belief-update map F_t . The coordinator therefore faces a finite-horizon fully
 779 observed MDP on belief state b_t , with action equal to the choice of prescription Γ_t .

780 **Backward induction.** Let $W_t(b)$ denote the optimal coordinator value from stage t onward. Since
 781 both the belief space and the prescription space are finite-dimensional simplices over finite alphabets,
 782 standard backward induction applies:

$$W_t(b) = \max_{\Gamma_t} \mathbb{E}[r_t + W_{t+1}(b_{t+1}) \mid b_t = b, \Gamma_t],$$

783 with terminal condition $W_{H+1} \equiv 0$. Hence there exists an optimal measurable selector $\Gamma_t^*(b_t)$
 784 depending only on the current common belief.

785 **Decentralized implementation.** Given the coordinator prescription $\Gamma_t^*(b_t)$, teammate i executes

$$a_t^i \sim \gamma_t^{i,*}(\cdot \mid x_t^i).$$

786 This action depends only on the teammate's private information x_t^i and on the common belief
 787 summarized by the public history, so it is C4-compliant. Because the coordinator MDP and the
 788 original decentralized best-response problem induce the same controlled trajectory law, their optimal
 789 values agree. This proves that public belief is a sufficient team-side state variable and that an optimal
 790 decentralized response can be implemented through prescriptions measurable with respect to b_t alone.
 791 $\square$

792 **Reading in this paper.** The proposition does not claim that b_t is tractable to represent exactly
 793 in Doudizhu or in replay-based execution. Its role is structural: once C4 is imposed, public belief
 794 becomes the canonical sufficient statistic, so critic-side belief support and public-signal shaping are
 795 not ad hoc tricks but approximations to the same reduced control object.

796 A.12 Proof of Proposition 2

797 **Finite-space convention for this subsection.** Assume a finite horizon H , finite teammate-private
 798 alphabets, and finite public-signal alphabets for both Y' and Y . Let $\Pi^{\text{Dec}}(Y')$ and $\Pi^{\text{Dec}}(Y)$ denote
 799 the C4-compliant decentralized team policy classes whose agent- i decision rules may depend on
 800 $(x_t^i, Y'_{0:t})$ or $(x_t^i, Y_{0:t})$, respectively. Because all alphabets are finite, every policy can be identified
 801 with a collection of conditional action distributions, and every required conditional law exists as an
 802 ordinary stochastic kernel on a finite set.

803 **Public garbling assumption.** For each stage t , suppose there exists a publicly known stochastic
 804 kernel

$$\kappa_t(y_t \mid y'_{0:t}, y_{0:t-1})$$

805 such that, given a realization of the richer public trace $Y'_{0:t}$, the coarser trace Y_t is sampled according
 806 to κ_t , and any extra randomization used by κ_t is publicly shared. Equivalently, conditional on $Y'_{0:t}$
 807 and the past coarse trace $Y_{0:t-1}$, the law of Y_t depends on no teammate-private variable.

808 **Policy lifting.** Fix any coarse-signal decentralized team policy $\pi \in \Pi^{\text{Dec}}(Y)$. We construct a lifted
 809 policy $\pi^\uparrow \in \Pi^{\text{Dec}}(Y')$ as follows. At each stage t , all teammates observe the same richer public
 810 trace $Y'_{0:t}$. Using the shared public randomization in the garbling kernels $(\kappa_\ell)_{\ell \leq t}$, they recursively
 811 generate a synthetic coarse trace

$$\tilde{Y}_{0:t} \sim \mathbb{P}(Y_{0:t} \mid Y'_{0:t})$$

812 with exactly the same conditional law induced by the public garbling process. Agent i then applies
 813 the original coarse-signal decision rule of π to $(x_t^i, \tilde{Y}_{0:t})$. Since $\tilde{Y}_{0:t}$ is a function only of public
 814 information and shared public randomization, $\pi^\uparrow$ remains C4-compliant and belongs to $\Pi^{\text{Dec}}(Y')$.

815 **Trajectory-law equality.** We claim that $\pi^\uparrow$ reproduces under Y' the same joint law over states,
 816 actions, rewards, and synthetic coarse public traces that π induces under Y . This is proved by
 817 induction on t . At $t = 0$, the synthetic trace $\tilde{Y}_0$ is sampled from the same kernel that generates
 818 Y_0 from Y'_0 , and the lifted agent actions use exactly the same conditional action law as π after
 819 substituting Y_0 by $\tilde{Y}_0$. Assume the claim holds through stage t . Because the environment dynamics,
 820 fixed adversary policy π_K , and lifted team action law at stage t agree with those of the coarse system
 821 after replacing $Y_{0:t}$ by $\tilde{Y}_{0:t}$, the joint law of the next state and reward also agrees. Applying the public
 822 garbling kernel κ_{t+1} to the newly observed richer trace then yields a next synthetic coarse symbol
 823 $\tilde{Y}_{t+1}$ with the correct conditional law. Thus the claim propagates to stage $t + 1$.

824 **Value comparison.** By the trajectory-law equality, the expected return of $\pi^\uparrow$ against the fixed
 825 adversary π_K under the richer signal Y' equals the expected return of π under the coarser signal Y .
 826 Since $\pi \in \Pi^{\text{Dec}}(Y)$ was arbitrary, every coarse-signal policy has a richer-signal lift with identical
 827 value. Hence

$$\sup_{\pi' \in \Pi^{\text{Dec}}(Y')} V^{\pi', \pi_K} \geq \sup_{\pi \in \Pi^{\text{Dec}}(Y)} V^{\pi, \pi_K},$$

828 which is exactly

$$V_{\text{Dec}}^*(Y'; \pi_K) \geq V_{\text{Dec}}^*(Y; \pi_K).$$

829

□

830 **Reading in this paper.** The proposition is intentionally local to the legal public trace and to a fixed
 831 adversary. It does not say that every refinement is cheap to learn or that all added public bits are
 832 strategically useful in finite training. It says only that, once the extra signal is legal and publicly
 833 shared, it weakly enlarges what a C4-compliant team could in principle achieve. This is the structural
 834 justification for treating Module B as public-signal refinement rather than as hidden communication.

835 A.13 Proof of Proposition 3

836 **Finite-space and measure convention for this subsection.** Assume a finite horizon, a finite
 837 summary alphabet $\mathcal{Z}$ with $|\mathcal{Z}| \geq 2$, and a finite legal-information alphabet for L_t^j . Fix the C4-
 838 compliant team policy π and the fixed adversary policy from the proposition, and write $\mathbb{P}^\pi$ and $\mathbb{E}^\pi$
 839 for the induced trajectory law and expectation. All conditional entropies, mutual informations, and
 840 error probabilities in this proof are computed under $\mathbb{P}^\pi(\cdot | C_t)$. Since the alphabets are finite and
 841 $\rho_{\text{crit}}^\pi = \mathbb{P}^\pi(C_t) > 0$, the conditional mutual information $I_\pi(Z_t; L_t^j | C_t)$ is well defined, the decoder
 842 infimum is attained by a minimizer, and all conditional expectations below are finite sums.

843 **Legal decoder and action-relevance assumption.** The proposition applies after Z_t has been
 844 reduced to an action-relevant summary: if two raw hidden summaries induce the same centralized
 845 continuation, they are first merged into the same value of Z_t . Under this convention, a mistake in
 846 identifying Z_t is a mistake in identifying the centralized continuation class that matters for value.
 847 The proposition then assumes that the continuation chosen by π induces a legal decoder $\hat{Z}_\pi(L_t^j)$ and
 848 that the continuation-value gap satisfies the pointwise margin condition

$$D_\pi \geq \lambda \mathbf{1}\{\hat{Z}_\pi(L_t^j) \neq Z_t\} \quad \text{on } C_t.$$

849 This condition is the explicit formal version of the informal phrase that a decoding mistake costs at
 850 least λ in continuation value.

851 Because $\hat{Z}_\pi$ is measurable with respect to L_t^j , it is one admissible legal decoder. Therefore, if

$$P_e^\pi := \inf_{\hat{Z}} \mathbb{P}^\pi(\hat{Z}(L_t^j) \neq Z_t | C_t),$$

852 where the infimum ranges over all measurable legal decoders $\hat{Z} : L_t^j \mapsto \mathcal{Z}$, then

$$\mathbb{P}^\pi(\hat{Z}_\pi(L_t^j) \neq Z_t | C_t) \geq P_e^\pi.$$

853 **Continuation-loss lower bound.** Using the margin condition and the definition of $\varepsilon_{\text{coord}}^\pi$,

$$\begin{aligned} \varepsilon_{\text{coord}}^\pi &= \mathbb{E}^\pi[D_\pi \mathbf{1}_{C_t}] \\ &\geq \lambda \mathbb{P}^\pi(\hat{Z}_\pi(L_t^j) \neq Z_t, C_t) \\ &= \lambda \rho_{\text{crit}}^\pi \mathbb{P}^\pi(\hat{Z}_\pi(L_t^j) \neq Z_t | C_t) \\ &\geq \lambda \rho_{\text{crit}}^\pi P_e^\pi. \end{aligned}$$

854 **Conditional Fano step.** Apply Fano's inequality to the problem of reconstructing Z_t from L_t^j
 855 under the conditional law $\mathbb{P}^\pi(\cdot | C_t)$. For the optimal decoder error P_e^π ,

$$H_\pi(Z_t | L_t^j, C_t) \leq h_2(P_e^\pi) + P_e^\pi \log_2(|\mathcal{Z}| - 1),$$

where $h_2(p) := -p \log_2 p - (1-p) \log_2 (1-p)$ is the binary entropy. Using $h_2(P_e^\pi) \leq 1$ and $\log_2(|\mathcal{Z}| - 1) \leq \log_2 |\mathcal{Z}|$ gives

$$H_\pi(Z_t | L_t^j, C_t) \leq 1 + P_e^\pi \log_2 |\mathcal{Z}|.$$

Since

$$H_\pi(Z_t | L_t^j, C_t) = H_\pi(Z_t | C_t) - I_\pi(Z_t; L_t^j | C_t),$$

rearranging gives

$$P_e^\pi \geq \left[\frac{H_\pi(Z_t | C_t) - I_\pi(Z_t; L_t^j | C_t) - 1}{\log_2 |\mathcal{Z}|} \right]_+.$$

Substituting this into the previous display gives

$$\varepsilon_{\text{coord}}^\pi \geq \lambda \rho_{\text{crit}}^\pi \left[\frac{H_\pi(Z_t | C_t) - I_\pi(Z_t; L_t^j | C_t) - 1}{\log_2 |\mathcal{Z}|} \right]_+,$$

which is the claimed bound. $\square$

Reading in this paper. This bound isolates what Module B can and cannot do under a fixed evaluation occupancy and a fixed action-relevant summary. If legal teammate-side information already contains enough bits about Z_t , the lower bound can become small. If not, and if the explicit margin condition holds for the continuation classes indexed by Z_t , then the C4-compliant policy cannot drive $\varepsilon_{\text{coord}}^\pi$ to zero merely by better optimization. The theorem therefore turns the legal public channel from a heuristic communication story into a conditional information-bottleneck certificate.

A.14 Legal-capacity lower bound and probe protocol

Definition. Fix a coordination-critical event C_t , a sender-private summary Z_t , and receiver-local information x_t^j . The legal coordination capacity of the public trace for receiver j is

$$C_{\text{legal}} := I(Z_t; h_t^{\text{pub}} | x_t^j, C_t).$$

This quantity measures how much teammate-relevant information the legal public history carries beyond what receiver j already knows from its own local state.

Proposition 7 (Probe-based lower bound on legal coordination capacity). *Fix a coordination-critical event C_t , a sender-private summary Z_t , and receiver-local information x_t^j . Let*

$$C_{\text{legal}} := I(Z_t; h_t^{\text{pub}} | x_t^j, C_t).$$

If $q_\psi(z | x_t^j, h_t^{\text{pub}}, C_t)$ is any held-out receiver probe with cross-entropy

$$\mathcal{L}_{\text{rec}}^{\text{held}} := \mathbb{E} \left[-\log_2 q_\psi(Z_t | x_t^j, h_t^{\text{pub}}, C_t) \right],$$

then

$$C_{\text{legal}} \geq H(Z_t | x_t^j, C_t) - \mathcal{L}_{\text{rec}}^{\text{held}}.$$

Proof. Let $q_\psi(z | x_t^j, h_t^{\text{pub}}, C_t)$ be any held-out receiver probe, and define

$$\mathcal{L}_{\text{rec}}^{\text{held}} := \mathbb{E} \left[-\log_2 q_\psi(Z_t | x_t^j, h_t^{\text{pub}}, C_t) \right].$$

Because cross-entropy upper-bounds the true conditional entropy,

$$\mathcal{L}_{\text{rec}}^{\text{held}} \geq H(Z_t | x_t^j, h_t^{\text{pub}}, C_t).$$

Therefore

$$\begin{aligned} C_{\text{legal}} &= H(Z_t | x_t^j, C_t) - H(Z_t | x_t^j, h_t^{\text{pub}}, C_t) \\ &\geq H(Z_t | x_t^j, C_t) - \mathcal{L}_{\text{rec}}^{\text{held}}. \end{aligned}$$

$\square$

881 **Reading in this paper.** This is the bound stated in the main text. It is a lower bound on legal
 882 coordination informativeness, not a claim that the probe recovers the exact Shannon capacity of the
 883 induced public channel.

884 **Why this bound is a natural fit for Module B.** The lower bound is tailored to the present paper
 885 because it reuses the sender/receiver auxiliary structure that Module B already trains, rather than
 886 introducing a separate high-variance mutual-information estimator. In particular, the relevant receiver-
 887 side quantity is just the held-out cross-entropy or BCE of reconstructing the sender summary Z_t from
 888 legal public traces. This makes the probe lightweight, domain-matched, and directly auditable from
 889 frozen trajectories. In other words, the paper is not inventing an unrelated diagnostic; it is converting
 890 the existing auxiliary decoding loss into a lower bound on legal coordination information.

891 **Comparing Base and Module B under the same target definition.** If Base and Module B are
 892 evaluated on the same held-out distribution, the same coordination-critical slice C_t , and the same
 893 sender-summary definition Z_t , then the lower-bound comparison becomes especially transparent. Let
 894 $\mathcal{L}_{\text{rec}}^{\text{Base}}$ denote the held-out receiver loss for the Base model and let $\mathcal{L}_{\text{rec}}^{\text{B}}$ denote the corresponding
 895 loss for the Module-B-equipped model. Since the entropy term $H(Z_t | x_t^j, C_t)$ is shared under the
 896 matched comparison protocol, the improvement in the lower bound is at least

$$\Delta C_{\text{legal}}^{\text{lb}} := [H(Z_t | x_t^j, C_t) - \mathcal{L}_{\text{rec}}^{\text{B}}] - [H(Z_t | x_t^j, C_t) - \mathcal{L}_{\text{rec}}^{\text{Base}}] = \mathcal{L}_{\text{rec}}^{\text{Base}} - \mathcal{L}_{\text{rec}}^{\text{B}}.$$

897 If logarithms are reported in bits rather than nats, the usual log-base conversion applies. The
 898 interpretation is direct: lower receiver BCE / cross-entropy means that sender-side summaries are
 899 more decodable from legal public traces, so the lower bound on legal coordination capacity is
 900 correspondingly higher.

901 **No-public control.** To isolate information provided by the public trace rather than by receiver-local
 902 priors, we also evaluate a control probe

$$q_{\psi}^{\text{nopub}}(z | x_t^j, C_t)$$

903 with held-out loss

$$\mathcal{L}_{\text{nopub}}^{\text{held}} := \mathbb{E} \left[-\log_2 q_{\psi}^{\text{nopub}}(Z_t | x_t^j, C_t) \right].$$

904 The gap $\mathcal{L}_{\text{nopub}}^{\text{held}} - \mathcal{L}_{\text{rec}}^{\text{held}}$ is the probe-side gain attributable to legal public history beyond receiver-local
 905 information alone.

906 **Held-out comparison protocol used in the paper.** The probe is run with all trained policies
 907 frozen and evaluated on matched held-out trajectories under the same domain-specific scripts used
 908 by the main experiments. We compare Base, Base + Module A + Module B, and Full on the same
 909 coordination-critical slice C_t . In Doudizhu, Z_t is the Module-B hand-strength / remaining-hand target
 910 used by the sender/receiver auxiliaries; in multi-contract execution, Z_t is the sender urgency bin. The
 911 public probe consumes $(x_t^j, h_t^{\text{pub}})$, the control probe consumes x_t^j alone, and the reported statistics
 912 are held-out receiver cross-entropy, decode accuracy, the lower-bound proxy $H(Z_t | x_t^j, C_t) - \mathcal{L}_{\text{rec}}^{\text{held}}$,
 913 and the no-public gap $\mathcal{L}_{\text{nopub}}^{\text{held}} - \mathcal{L}_{\text{rec}}^{\text{held}}$. Empirical entropies such as $H(Z_t | x_t^j, C_t)$ are estimated
 914 from the same held-out split by Monte Carlo counts over the finite summary bins.

915 A.15 Proof of Proposition 8

916 **Proposition 8** (Occupancy-average action-information in the legal public trace). *Assume (PC1) a_t^i*
 917 *enters teammate j 's next observation o_{t+1}^j as a deterministic component of public history; (PC2)*
 918 *under the evaluated occupancy measure d^{π} over sender histories τ , there exist $p_{\min} \in (0, 1)$ and*

$$\rho_{\text{nd}} := \Pr_{\tau \sim d^{\pi}} [H(\pi_i(\cdot | \tau)) \geq h_2(p_{\min})] > 0;$$

919 *and (PC3) the support of $\pi_i(\cdot | \tau)$ is finite for d^{π} -almost every τ . Then*

$$\mathbb{E}_{\tau \sim d^{\pi}} [I(a_t^i; o_{t+1}^j | \tau_t^i = \tau)] = \mathbb{E}_{\tau \sim d^{\pi}} [H(\pi_i(\cdot | \tau))],$$

920 *and therefore*

$$\rho_{\text{nd}} h_2(p_{\min}) \leq \mathbb{E}_{\tau \sim d^{\pi}} [I(a_t^i; o_{t+1}^j | \tau_t^i = \tau)] \leq \mathbb{E}_{\tau \sim d^{\pi}} [\log_2 |\text{supp}(\pi_i(\cdot | \tau))|].$$

921 **Proof.** Under PC1, teammate j 's next observation o_{t+1}^j contains a_t^i as a deterministic component
 922 of the public history. Under PC2, there exists an occupancy measure d^π over sender histories τ , a
 923 constant $p_{\min} \in (0, 1)$, and a non-degenerate set

$$G := \{\tau : H(\pi_i(\cdot | \tau)) \geq h_2(p_{\min})\}$$

924 with occupancy mass

$$\rho_{\text{nd}} := \Pr_{\tau \sim d^\pi} [\tau \in G] > 0.$$

925 Under PC3, the support of $\pi_i(\cdot | \tau)$ is finite for d^π -almost every τ .

926 **Domain verification of the structural conditions.** The cross-domain comparison in the paper
 927 uses PC1 and PC3 as structural task properties, while PC2 is an occupancy-level non-degeneracy
 928 condition checked on the evaluated sender history distribution. This distinction matters because the
 929 two domains use different base learners. In Doudizhu, the deployed stack is DMC-style and may be
 930 deterministic at many realized local histories, yet the evaluation occupancy still places positive mass
 931 on histories at which bidding or high-commitment play is genuinely non-degenerate. In multi-contract
 932 execution, the PPO-style actor often satisfies the same condition more directly through its stochastic
 933 action law, though the proposition no longer requires every visited history to be stochastic. In both
 934 domains, bids, played cards, submitted orders, and trades are revealed immediately, so teammate
 935 actions enter the public trace after execution. The domains therefore differ in semantics, policy
 936 parameterization, and alphabet richness, but they share the legal public-channel structure required by
 937 Proposition 8. Whenever the evaluated history occupancy contains a positive-mass non-degenerate
 938 subset, the proposition applies.

939 **Step 1: conditional exact equality.** Fix any sender history τ in the support of d^π , and let $A_\tau \sim$
 940 $\pi_i(\cdot | \tau)$ denote the sender action random variable at that history. Because A_τ is a deterministic
 941 component of teammate j 's next observation, we have

$$H(A_\tau | o_{t+1}^j, \tau_t^i = \tau) = 0.$$

942 Therefore

$$I(A_\tau; o_{t+1}^j | \tau_t^i = \tau) = H(A_\tau) - H(A_\tau | o_{t+1}^j, \tau_t^i = \tau) = H(\pi_i(\cdot | \tau)).$$

943 Taking expectation over $\tau \sim d^\pi$ proves the exact identity in Proposition 8.

944 **Step 2: positive lower bound under occupancy-level non-degeneracy.** By definition of G , every
 945 $\tau \in G$ satisfies

$$H(\pi_i(\cdot | \tau)) \geq h_2(p_{\min}).$$

946 Since $p_{\min} \in (0, 1)$, $h_2(p_{\min}) > 0$. Therefore

$$\mathbb{E}_{\tau \sim d^\pi} [H(\pi_i(\cdot | \tau))] \geq \mathbb{E}_{\tau \sim d^\pi} [\mathbf{1}[\tau \in G] H(\pi_i(\cdot | \tau))] \geq \rho_{\text{nd}} h_2(p_{\min}) > 0.$$

947 Combining this with Step 1 yields the occupancy-average lower bound in the proposition.

948 **Step 3: occupancy-average alphabet upper bound.** For each τ , if $\pi_i(\cdot | \tau)$ has finite support,
 949 then Shannon entropy is maximized by the uniform distribution on that support. Hence

$$H(\pi_i(\cdot | \tau)) \leq \log_2 |\text{supp}(\pi_i(\cdot | \tau))|.$$

950 Applying Step 1 and taking expectation again yields

$$\mathbb{E}_{\tau \sim d^\pi} [I(a_t^i; o_{t+1}^j | \tau_t^i = \tau)] \leq \mathbb{E}_{\tau \sim d^\pi} [\log_2 |\text{supp}(\pi_i(\cdot | \tau))|].$$

951 This completes the proof.

952 **What the proposition does and does not prove.** The proposition shows that the legal public trace
 953 is information-bearing at the action-observability level whenever teammate actions are explicitly
 954 revealed and used non-degenerately. It does *not* claim that this quantity alone equals private-state
 955 coordination capacity or that Module B recovers all coordination value. Those stronger questions are
 956 handled separately by the legal-capacity lower bound, the explicit-latent upper bound, the public-
 957 channel intervention, and the cross-domain recovery experiments.

958 **Corollary 1** (Sender-private informativeness of public actions). *Let x_t^i denote the sender-private*
 959 *component of teammate i 's information at time t , so that $\tau_t^i = (h_t^{\text{pub}}, x_t^i)$ with h_t^{pub} the public history.*
 960 *Suppose there exists a positive-measure set of public histories such that, conditioned on each such*
 961 *history, there are sender-private states x, x' with*

$$\Pr(x_t^i = x \mid h_t^{\text{pub}}) > 0, \quad \Pr(x_t^i = x' \mid h_t^{\text{pub}}) > 0,$$

962 *and the conditional action distribution depends nontrivially on that private state:*

$$\pi_i(\cdot \mid h_t^{\text{pub}}, x) \neq \pi_i(\cdot \mid h_t^{\text{pub}}, x').$$

963 *Then*

$$I(x_t^i; a_t^i \mid h_t^{\text{pub}}) > 0,$$

964 *and, under PC1,*

$$I(x_t^i; o_{t+1}^j \mid h_t^{\text{pub}}) \geq I(x_t^i; a_t^i \mid h_t^{\text{pub}}) > 0.$$

965 *Thus whenever the sender policy uses the sender's private information in a domain-relevant way, the*
 966 *legal public action reveals strictly positive teammate-relevant information about that private state.*

967 **Proof.** If the conditional action law varies with x_t^i on a positive-measure set of public histories and
 968 the two witness states x, x' both have positive conditional probability given those histories, then a_t^i
 969 is not conditionally independent of x_t^i given h_t^{pub} , hence $I(x_t^i; a_t^i \mid h_t^{\text{pub}}) > 0$. Under PC1, a_t^i is a
 970 deterministic function of o_{t+1}^j , so the data-processing inequality gives

$$I(x_t^i; o_{t+1}^j \mid h_t^{\text{pub}}) \geq I(x_t^i; a_t^i \mid h_t^{\text{pub}}).$$

971 Combining the two statements yields the corollary. □

972 **Interpretation.** Corollary 1 is the domain-level bridge from public action entropy to teammate-
 973 useful signaling. In Doudizhu, high-commitment plays and bids can reveal private hand strength
 974 or control resources. In multi-contract execution, action templates can reveal urgency, inventory
 975 pressure, or preferred execution cadence. The result does not say that every revealed bit is strategically
 976 useful; it says that useful sender-private information can legally enter the public trace whenever the
 977 sender policy depends on that private state.

978 **Corollary 2** (Well-posedness and domain-agnostic validity of Module B). *Under PC1–PC3, the*
 979 *public trace reveals a strictly positive occupancy-averaged amount of sender-action information*
 980 *to teammates. Hence the sender/receiver auxiliaries in Module B are not vacuous: there exists a*
 981 *legal public channel whose informativeness can, in principle, be increased by learning. Because*
 982 *Doudizhu and multi-contract execution share the same structural public-channel conditions, the*
 983 *same argument applies in both domains once the evaluated sender history occupancy contains a*
 984 *positive-mass non-degenerate subset. In this precise sense, Module B is domain-agnostic within*
 985 *the AT-Dec-POSG framework: it relies on legality and informativeness of the public trace, not on*
 986 *whether the underlying learner is a DMC-style Q-network or a PPO-style actor-critic.*

987 **Proof.** The non-vacuity claim is immediate from Proposition 8, which gives a strictly positive
 988 occupancy-averaged lower bound whenever PC1–PC3 hold. The domain-agnostic reading then
 989 follows because those conditions are stated at the level of public-channel legality, action revelation,
 990 and occupancy-level non-degeneracy rather than at the level of any specific learner architecture. □

991 **Corollary 3** (Cumulative public-channel budget in expectation). *Over a horizon of T informative*
 992 *decision steps, let d_t^π denote the sender-history occupancy at step t , and let $(\rho_{\text{nd},t}, p_{\text{min},t})$ be any*
 993 *occupancy-level non-degeneracy pair satisfying PC2 at that step. Then the total expected revealed*

994 *action information satisfies*

$$\begin{aligned}
& \sum_{t=1}^T \rho_{\text{nd},t} h_2(p_{\min,t}) \\
& \leq \sum_{t=1}^T \mathbb{E}_{\tau \sim d_t^\pi} \left[I(a_t^i; o_{t+1}^j \mid \tau_t^i = \tau) \right] \\
& = \sum_{t=1}^T \mathbb{E}_{\tau \sim d_t^\pi} [H(\pi_i(\cdot \mid \tau))] \\
& \leq \sum_{t=1}^T \mathbb{E}_{\tau \sim d_t^\pi} [\log_2 |\text{supp}(\pi_i(\cdot \mid \tau))|].
\end{aligned}$$

995 *Thus the legal signaling budget scales at most linearly with the number of informative decision points,*
996 *but its positive lower bound now depends on the occupancy mass of non-degenerate histories rather*
997 *than on every visited history being stochastic.*

998 **Proof.** Apply Proposition 8 at each informative decision step t , then sum the resulting lower and
999 upper bounds over $t = 1, \dots, T$. Linearity of expectation gives the middle identity, and the displayed
1000 inequalities follow term by term. $\square$

1001 **Proposition 9** (Assumption-based bridge from legal signaling budget to recovery ratio). *Let*

$$B_T = \sum_{t=1}^T \mathbb{E}_{\tau \sim d_t^\pi} \left[I(a_t^i; o_{t+1}^j \mid \tau_t^i = \tau) \right]$$

1002 *denote the cumulative legal public-channel budget over the informative decision steps of an episode,*
1003 *and assume $\Delta_{\text{coord}} > 0$. Suppose the experimental proxy is calibrated exactly for the evaluated*
1004 *protocol, so that*

$$\varepsilon_B = \Delta_{\text{coord}}(1 - \rho_{\text{coord}}).$$

1005 *Then any bound $\varepsilon_B \leq \bar{\varepsilon}_B$ implies*

$$\rho_{\text{coord}} \geq 1 - \frac{\bar{\varepsilon}_B}{\Delta_{\text{coord}}}.$$

1006 *In particular, if there exists a domain-dependent constant $\eta > 0$ such that*

$$\varepsilon_B \leq \Delta_{\text{coord}} \exp(-B_T/\eta),$$

1007 *then*

$$\begin{aligned}
\rho_{\text{coord}} & \geq 1 - \exp(-B_T/\eta) \\
& \geq 1 - \exp\left(-\frac{\sum_{t=1}^T \rho_{\text{nd},t} h_2(p_{\min,t})}{\eta}\right).
\end{aligned}$$

1008 **Proof.** Under the calibration relation,

$$\rho_{\text{coord}} = 1 - \frac{\varepsilon_B}{\Delta_{\text{coord}}}.$$

1009 So any upper bound $\varepsilon_B \leq \bar{\varepsilon}_B$ yields the first inequality immediately. Applying the exponential residue
1010 model and then Corollary 3 gives the stated lower bound in terms of both B_T and $\sum_t \rho_{\text{nd},t} h_2(p_{\min,t})$.

1011 **Interpretation.** This proposition is intentionally conditional: Proposition 8 alone proves only
1012 that the legal channel is non-vacuous in occupancy-average, whereas Proposition 9 adds one extra
1013 calibration step from bits to recoverable value. Its use in the paper is therefore predictive rather
1014 than absolute. If two domains share a comparable bits-to-value scale η , then larger cumulative legal
1015 public-channel budgets should support larger recovery ratios ρ_{coord} .

1016 **Cross-domain reading.** These public-channel results explain why Doudizhu and multi-contract
 1017 execution are both suitable testbeds for Module B. In Doudizhu, the public channel starts already in
 1018 bidding, where pass / bid-level actions form a low-bandwidth scalar alphabet, and then widens in
 1019 card play, where combinatorial action types help teammates infer hand strength or control resources.
 1020 In multi-contract execution, aggressiveness and size templates induce a smaller but still multi-way
 1021 public action alphabet whose public realization helps teammates infer execution cadence or urgency.
 1022 Doudizhu therefore offers a richer legal alphabet and potentially larger raw signaling budget, whereas
 1023 execution uses a coarser alphabet but needs to encode only lower-bandwidth intentions such as
 1024 urgency or pacing. The domains need not share the same absolute capacity; what matters is that both
 1025 expose a nontrivial finite legal public channel under C4, so the same lower-bound logic applies in
 1026 each case.

1027 A.16 Approximate-equilibrium decomposition details

1028 A.16.1 Why error decomposition rather than convergence

1029 Full convergence to an exact Nash equilibrium under deep function approximation is an open problem
 1030 in adversarial MARL. We therefore do not claim global convergence. Instead, this appendix proves a
 1031 weaker but more actionable statement: the learned pair is near-equilibrium whenever three module-
 1032 specific residuals are small. This is stronger than Proposition 6 alone, because it connects the final
 1033 team-side and adversary-side deviation gaps directly to measurable errors associated with Modules A,
 1034 B, and C.

Module	Residual term	Interpretation	Controlled by
Module A	ε_A	critic mismatch	distillation / calibration quality
Module B	ε_B	unrecovered legal coordina- tion residue	$\Delta_{\text{coord}}, \Delta_{\text{unrec}}, \rho_{\text{coord}}$
Module C	ε_C	league-side adversary cov- erage gap	league diversity and held-out ex- ploiter

1036 A.16.2 Definitions

1037 For the residual decomposition in this appendix, fix an evaluation initial-state distribution ξ and write
 1038 $V^{\pi_{\mathcal{M}}, \pi_{\mathcal{K}}}$ for the expected discounted return under $s_0 \sim \xi$. The earlier pointwise coordination-loss
 1039 propositions remain state-wise; the distributional convention is introduced here so that the equilibrium
 1040 statement and the scalar residuals live on the same value scale.

1041 **Definition 5** (AT-Dec-POSG ε -equilibrium). *The trained pair $(\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}})$ is an ε -equilibrium under ξ*
 1042 *if*

$$V^{\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} \geq \sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}} V^{\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} - \varepsilon,$$

1043 *and*

$$V^{\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} \leq \inf_{\pi_{\mathcal{K}}} V^{\hat{\pi}_{\mathcal{M}}, \pi_{\mathcal{K}}} + \varepsilon.$$

1044 **Definition 6** (Module-wise residuals). *For the trained pair $(\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}})$, let $\tilde{V}(\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}})$ denote the*
 1045 *approximate team-side objective optimized by Modules A and B. Define*

$$\varepsilon_A = \sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}} \left| \tilde{V}(\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}) - V^{\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} \right|,$$

1046

$$\varepsilon_B = \sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}} \tilde{V}(\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}) - \tilde{V}(\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}),$$

1047 *and*

$$\varepsilon_C = V^{\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} - \inf_{\pi_{\mathcal{K}}} V^{\hat{\pi}_{\mathcal{M}}, \pi_{\mathcal{K}}}.$$

1048 **Unit consistency.** All three residuals ε_A , ε_B , and ε_C are measured in the same units as V , namely
 1049 expected cumulative reward. The aggregation

$$\varepsilon = \max\{2\varepsilon_A + \varepsilon_B, \varepsilon_C\}$$

1050 is therefore dimensionally well-defined rather than a sum of heterogeneous diagnostics.

1051 A.16.3 Residual identities and strengthened calibrations

1052 The bounds A1–A3 below are not independent theorem assumptions: they are the same inequalities
 1053 obtained by unpacking Definition 6’s module-wise residuals. They are recorded explicitly only
 1054 because the later proof repeatedly invokes them as reusable shorthand. The genuinely additional
 1055 modeling conditions are the strengthened calibrations A1’–A3’.

1056 **Residual identity A1 (critic residual control).** The approximate objective used by Modules A and
 1057 B satisfies the uniform error bound

$$\left| \tilde{V}(\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}) - V^{\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} \right| \leq \varepsilon_A \quad \text{for all } \pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}.$$

1058 **Limitation note.** A1 is not an extra theorem assumption; it simply restates the definition of
 ε_A as a reusable bound. In the paper, ε_A should still be interpreted through measured training
 residuals such as distillation loss, value calibration, or held-out Bellman-style diagnostics rather
 than presumed to vanish automatically.

1059 **Residual identity A2 (legal coordination residual).** The learned decentralized team policy is
 1060 ε_B -suboptimal with respect to the approximate team-side objective:

$$\sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}} \tilde{V}(\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}) - \tilde{V}(\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}) \leq \varepsilon_B.$$

1061 Operationally, the most interpretable proxy is the unrecovered legal coordination residue

$$\varepsilon_B \approx \Delta_{\text{unrec}} = \Delta_{\text{coord}}(1 - \rho_{\text{coord}}).$$

1062 Here Δ_{coord} always denotes the total no-coordination-to-explicit gap and Δ_{unrec} the remaining
 1063 explicit-to-Module-B residue; the two symbols are not interchangeable.

1064 **Limitation note.** A2 is likewise a bookkeeping restatement of the definition of ε_B , not a claim
 that Module B is globally optimal in the decentralized class. Proposition 8 justifies why this
 term is non-vacuous under C4, but its actual size remains an empirical question.

1065 **Residual identity A3 (league-side adversary coverage).** The league-trained adversary leaves a
 1066 bounded residual to the best adversary response:

$$V^{\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} - \inf_{\pi_{\mathcal{K}}} V^{\hat{\pi}_{\mathcal{M}}, \pi_{\mathcal{K}}} \leq \varepsilon_C.$$

1067 **Limitation note.** A3 records the residual ε_C in inequality form so it can enter the decomposition
 proof directly. Interpreting that residual still requires population-robustness evaluation, held-out
 exploiters, and league diversity diagnostics rather than raw pool cardinality alone.

1068 **Strengthened calibration A1’ (distillation-to-value calibration).** On top of residual identity A1,
 1069 suppose the final Module-A critic admits the explicit calibration

$$2\varepsilon_A \leq \sqrt{2\lambda_1 L_{\text{distill}}^*}$$

1070 for some task-dependent constant $\lambda_1 > 0$ and measured terminal distillation loss L_{distill}^* . Here λ_1
 1071 absorbs any task-specific scaling between the reported loss and value units; in particular, if L_{distill}^* is
 1072 the MSE on return or value targets directly, no extra $(1 - \gamma)^{-2}$ normalization is introduced. This is
 1073 the closed-form Module-A term used in the calibrated version of Proposition 10.

1074 **Limitation note.** A1’ is stronger than A1: it asks not only for a bounded critic residual, but also
 for an explicit conversion from the measured distillation loss into the two-sided deviation penalty
 $2\varepsilon_A$. In the paper this should be read as a calibrated sufficient condition, not as a universal
 theorem for deep critics.

1075 **Strengthened calibration A2' (linear public-channel recovery rate).** Let $\Delta_{\text{recover}} =$
 1076 $\Delta_{\text{coord}}\rho_{\text{coord}}$ denote the coordination value recovered by Module B under the evaluation proto-
 1077 col. Assume that, after fixing the paper's bits-to-value calibration, each of the T informative decision
 1078 steps contributes at least $\delta_G > 0$ units of recoverable coordination value until the total coordination
 1079 gap is saturated:

$$\Delta_{\text{recover}} \geq \min\{\Delta_{\text{coord}}, \delta_G T\}.$$

1080 Equivalently,

$$\rho_{\text{coord}} \geq \min\left\{1, \frac{\delta_G T}{\Delta_{\text{coord}}}\right\},$$

1081 and therefore

$$\varepsilon_B = \Delta_{\text{coord}}(1 - \rho_{\text{coord}}) \leq \max(0, \Delta_{\text{coord}} - \delta_G T).$$

1082 **Limitation note.** A2' is stronger than Proposition 9. Proposition 9 gives an assumption-based
 bridge from legal signaling budget to recovery, whereas A2' replaces that bridge by a linear
 per-step recovery floor δ_G so that the final theorem can be written in compact closed form.

1083 **Strengthened calibration A3' (finite-pool size-rate bound).** Assume the league-side coverage
 1084 residual also satisfies the explicit size-rate bound

$$\varepsilon_C \leq \frac{\text{diam}(V)}{|\Pi_{\mathcal{K}}^L|}, \quad \text{diam}(V) = V_{\max} - V_{\min}.$$

1085 **Limitation note.** A3' compresses all questions of pool diversity, spacing, and exploiter quality
 into a simple $1/|\Pi_{\mathcal{K}}^L|$ rate. This is useful as a closed-form summary term, but it is stronger than
 the qualitative coverage bound A3 and should be interpreted as a measurable approximation, not
 as an automatic consequence of pool size alone.

1086 A.16.4 Error decomposition lemma

1087 **Lemma 2** (Error decomposition). *Under the module-wise bounds A1–A3,*

$$\sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}} V^{\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} - V^{\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} \leq 2\varepsilon_A + \varepsilon_B,$$

1088 and

$$V^{\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} - \inf_{\pi_{\mathcal{K}}} V^{\hat{\pi}_{\mathcal{M}}, \pi_{\mathcal{K}}} \leq \varepsilon_C.$$

1089 **Proof.** *Team side.* For any decentralized deviation $\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}$,

$$V^{\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} \leq \tilde{V}(\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}) + \varepsilon_A$$

1090 by A1, while

$$V^{\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} \geq \tilde{V}(\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}) - \varepsilon_A.$$

1091 Subtracting gives

$$V^{\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} - V^{\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} \leq \tilde{V}(\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}) - \tilde{V}(\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}) + 2\varepsilon_A.$$

1092 Taking the supremum over $\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}$ and applying A2 yields

$$\sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}} V^{\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} - V^{\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} \leq 2\varepsilon_A + \varepsilon_B.$$

1093 *Adversary side.* The second inequality is immediate from A3:

$$V^{\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} - \inf_{\pi_{\mathcal{K}}} V^{\hat{\pi}_{\mathcal{M}}, \pi_{\mathcal{K}}} \leq \varepsilon_C.$$

1094 □

1095 A.16.5 Main result: Proposition 10

1096 **Proposition 10** (Approximate-equilibrium decomposition). *Assume the module-wise bounds A1–A3*
 1097 *stated in Appendix A.16. Then the learned pair $(\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}})$ is an ε -equilibrium of the AT-Dec-POSG,*
 1098 *with*

$$\sup_{\pi_{\mathcal{M}} \in \Pi_{\mathcal{M}}^{\text{Dec}}} V^{\pi_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} - V^{\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} \leq 2\varepsilon_A + \varepsilon_B,$$

$$V^{\hat{\pi}_{\mathcal{M}}, \hat{\pi}_{\mathcal{K}}} - \inf_{\pi_{\mathcal{K}}} V^{\hat{\pi}_{\mathcal{M}}, \pi_{\mathcal{K}}} \leq \varepsilon_C.$$

1100 *Equivalently,*

$$\varepsilon = \max\{2\varepsilon_A + \varepsilon_B, \varepsilon_C\} \leq 2\varepsilon_A + \varepsilon_B + \varepsilon_C.$$

1101 **Restatement with strengthened calibrations.** If, in addition, the strengthened quantitative calibra-
 1102 tions A1'–A3' in Appendix A.16 hold, then

$$\varepsilon \leq \underbrace{\sqrt{2\lambda_1 L_{\text{distill}}^*}}_{\text{Module A}} + \underbrace{\max(0, \Delta_{\text{coord}} - \delta_G T)}_{\text{Module B}} + \underbrace{\frac{\text{diam}(V)}{|\Pi_{\mathcal{K}}^L|}}_{\text{Module C}}.$$

1103 Lemma 2 yields the structural decomposition in Proposition 10. Under the strengthened calibrations
 1104 A1'–A3', the same proposition also admits the following closed-form upper bound:

$$\varepsilon = \max\{2\varepsilon_A + \varepsilon_B, \varepsilon_C\} \leq 2\varepsilon_A + \varepsilon_B + \varepsilon_C \leq \sqrt{2\lambda_1 L_{\text{distill}}^*} + \max(0, \Delta_{\text{coord}} - \delta_G T) + \frac{\text{diam}(V)}{|\Pi_{\mathcal{K}}^L|}.$$

1105 The first inequality is elementary, the second is Lemma 2, and the final line applies A1'–A3' term
 1106 by term. This appendix supplies the assumptions and proof chain that make both the abstract
 1107 decomposition and the calibrated instantiation auditable. Proposition 8 is used only to interpret and
 1108 calibrate the legal-signaling term ε_B , not as a prerequisite for the theorem proof itself.

1109 **Asymptotic reading.** The decomposition has a clean directional interpretation:

- 1110 • $\sqrt{2\lambda_1 L_{\text{distill}}^*} \rightarrow 0$ as the Module-A critic's measured distillation loss shrinks.
- 1111 • $\max(0, \Delta_{\text{coord}} - \delta_G T) \rightarrow 0$ once Module B recovers enough legal public-channel value to
 1112 saturate the coordination gap.
- 1113 • $\text{diam}(V)/|\Pi_{\mathcal{K}}^L| \rightarrow 0$ as the league pool grows dense enough to cover more adversary responses.

1114 The paper does not claim that these limits hold automatically; it claims that these are the three
 1115 quantities that must become small if the final strategy pair is to be near-equilibrium.

1116 A.16.6 Connecting theory to experiments

1117 Proposition 10 is useful only if its terms are falsifiable. The experimental section is therefore designed
 1118 so that each residual corresponds to an observable diagnostic.

Residual	Meaning	Experimental proxy
ε_A	critic mismatch	final distillation loss, calibration error, or held-out value-prediction diagnostics for Module A
ε_B	unrecovered legal coordination residue	$\Delta_{\text{coord}}, \rho_{\text{coord}}$, explicit-latent upper bound, and public-channel intervention test
ε_C	adversary-side league coverage gap	held-out exploiter gap, population robustness, league diversity statistics, and time-to-threshold under league updates

1120 **Falsifiable prediction.** If all three residuals are small in both Doudizhu and multi-contract futures
 1121 execution, then Proposition 10 predicts that the learned pairs in both domains are near-equilibrium
 1122 in the AT-Dec-POSG sense. If any one residual remains large, the proposition no longer certifies
 1123 near-equilibrium, and the experiments should make that failure mode visible.

1124 **Calibrated reading for cross-domain comparison.** When the strengthened calibrations A1'–A3'
 1125 are used, the theorem can be read term by term in the same experimental units across domains:

Module	Closed-form term	Observable quantity	Comparative conclusion
Module A	$\sqrt{2\lambda_1 L_{\text{distill}}^*}$	final distillation loss or held-out value calibration	if small in both domains, critic mismatch is unlikely to explain the remaining differences
Module B	$\max(0, \Delta_{\text{coord}} - \delta_G T)$	coordination gap, recovery ratio, and public-channel intervention	if it shrinks in both domains despite different action semantics, the gain is explained by the shared legal public channel
Module C	$\text{diam}(V)/ \Pi_{\mathcal{K}}^L $	pool size, held-out exploiter gap, and robustness diagnostics	if it remains large in one domain, robustness coverage rather than coordination dominates the residual Nash error

1127 **Comparative conclusion.** Proposition 10 does more than introduce three residual labels. It predicts
 1128 that Doudizhu and multi-contract execution should exhibit the same *pattern of decomposition*: a critic
 1129 term, a legal-coordination term, and a league-coverage term. If the Module-B term shrinks in both
 1130 domains while the raw action spaces remain qualitatively different, then the cross-domain success of
 1131 the method is a structural consequence of AT-Dec-POSG rather than an empirical coincidence.

1132 A.16.7 Main statement dependency map

Statement	Role	Feeds into
Lem. 1	C4 creates a genuine coordination barrier	motivates Module B and the need for a coordination-residue term
Prop. 5	Module B stays C4-compliant at execution time	keeps the coordination mechanism inside the admissible policy class
Prop. 1	public belief is the reduced control state under fixed adversaries	motivates Module A as belief/state support
Prop. 2	richer legal public signals are weakly better than coarser ones	justifies public-signal refinement as a useful design objective
Prop. 3	coarse legal signals imply irreducible coordination residue	formalizes the part of coordination error that cannot be repaired away
Prop. 6	team-pool expansion is non-regressive when the adversary pool is fixed	supports the finite-league robustness role of Module C
Prop. 7	held-out decoding loss yields a legal-capacity lower bound	connects theorem-side informativeness to the probe reported in experiments
Prop. 10	module-wise residuals aggregate into a near-equilibrium statement	final bookkeeping conclusion

1134 **Reading the chain.** Lemma 1 explains why a gap exists at all. Propositions 1, 2, and 3 form the
 1135 main structural theory for what a legal public channel can represent, refine, and still fail to recover.
 1136 Proposition 5 keeps the mechanism inside C4, Proposition 6 explains the finite-league robustness role
 1137 of Module C, and Proposition 7 connects the theory to the held-out decoding probe. Proposition 10
 1138 then collects the module-wise residuals into the final near-equilibrium bookkeeping statement.

1139 A.16.8 Approximate-equilibrium reading

1140 **Practical reading of the decomposition result.** Under the public-channel conditions of Proposi-
 1141 tion 8 and the module-wise bounds A1–A3 in Appendix A.16, the three modules jointly produce an
 1142 ε -equilibrium whose gap decomposes as a critic residual, an unrecovered legal coordination residue,
 1143 and a league-coverage residual. This is a decomposition theorem rather than a full convergence
 1144 theorem, but it makes the end-to-end claim falsifiable because each term is paired with a concrete
 1145 experimental diagnostic.

1146 A.17 Proof sketch of the finance-side instantiation claim

1147 **Restatement of Example 2.** The main text records the replay-based multi-contract futures execution
 1148 benchmark as Example 2, with the market side represented through a task-level adversary abstraction.
 1149 This subsection gives the fuller construction and the longer caveats.

1150 Let the latent state collect the synchronized multi-contract market state together with all team-side
 1151 private execution states:

$$s_t = (\text{full books across contracts,} \\ \text{all outstanding team orders,} \\ \text{all inventory states,} \\ \text{episode schedule variables}).$$

1152 For execution agent i , define the local observation o_t^i as the projection of s_t onto contract-local book
 1153 features, its own outstanding orders, its own inventory target and remaining inventory, time-to-go, and
 1154 the public event tape. Let the shared team reward be the negative aggregate execution cost, including
 1155 implementation shortfall, inventory-risk penalties, and non-completion penalties.

1156 Under this construction, C3 holds because no single execution agent observes the full cross-contract
 1157 latent state or the private execution state of the rest of the team. C4 holds because agent i cannot
 1158 access teammate-private quantities such as hidden pending orders, internal signals, or inventory
 1159 targets; all teammate interaction must occur through the publicly visible order-flow history. Therefore
 1160 the replay-based multi-contract futures execution task is an instance of AT-Dec-POSG.

1161 **Adversary interpretation.** The market side is modeled as an aggregate adversarial agent whose
 1162 behavior is expressed through the replayed stream of limit-order placement, cancellation, and fills
 1163 generated by liquidity providers. Under this abstraction, $|\mathcal{K}| = 1$, and the zero-sum interpretation is
 1164 imposed at the task level: the execution team’s shortfall-minimization objective is directly opposed
 1165 to the liquidity side’s spread extraction and adverse-selection pressure. This abstraction does not
 1166 require a single explicitly parameterized opponent policy; it is enough that the team interacts with an
 1167 adversarial environment side through the public market dynamics.

1168 **Why futures execution rather than options trading.** The second domain is written as *multi-*
 1169 *contract futures execution* rather than the broader label *options trading* for a methodological reason
 1170 rather than a preference over asset classes. Futures execution yields the cleanest structural AT-Dec-
 1171 POSG instantiation: the team objective is naturally shared, the local/private state is easy to define
 1172 at the execution-agent level, the public channel is the market tape, and the adversarial side can be
 1173 modeled through aggregate liquidity response without introducing additional latent objects such as
 1174 volatility surfaces, Greeks, or hedging books. A multi-option setting is also plausible under the same
 1175 perspective, but it introduces richer private state and additional coupling constraints, so we reserve it
 1176 as a future extension instead of weakening the clarity of the present cross-domain comparison.

1177 A.18 Finance-side task definition

1178 The multi-contract execution benchmark is defined as an event-driven replay task over a fixed
 1179 universe of synchronized futures contracts. Each episode samples a historical window together
 1180 with an execution schedule specifying target inventory, horizon, and contract-specific participation
 1181 constraints. Agent i controls one contract or one portfolio slice and observes only contract-local state:
 1182 top-of-book or depth features, recent trade-flow summaries, its own outstanding orders, its remaining
 1183 inventory target, and time-to-go. The public state contains only market-visible events across the
 1184 contract set, namely orders, trades, and cancellations that would also be observable to any market
 1185 participant.

1186 The action space is a discrete set of execution templates indexed by aggressiveness and size bucket.
 1187 A passive action posts liquidity near the current quote, a neutral action tracks the mid or best quote
 1188 without consuming liquidity, and an aggressive action crosses the spread subject to size caps and
 1189 remaining inventory. Reward is shared at the team level and equals negative aggregate implementation
 1190 shortfall together with inventory-risk and non-completion penalties. The market-side opponent is not
 1191 introduced as an additional simulated neural agent; instead, adversarial pressure is represented by
 1192 the aggregate liquidity response implicit in the replayed order flow: the fill rates, price impacts, and
 1193 queue dynamics faced by the execution team are functions of how aggressively it participates, which
 1194 creates the same adversarial structure as an explicit opponent model at the task level.

1195 This construction gives a concrete interpretation of C3 and C4. C3 holds because each execution
 1196 agent sees only its own local state and the public tape, not the private target schedule or internal
 1197 latent state of its teammates. C4 holds because coordination may occur only through market-visible

actions that alter the shared public order-flow history. The execution benchmark tests structural C3/C4 transfer under replayed market-side pressure, not adaptation against a learned online adversary. Reproducibility is obtained by fixing the contract universe, split protocol, replay engine, preprocessing pipeline, action template set, and evaluation scripts for every reported experiment.

A.19 Module B experimental protocol

To make the C4-compliance evidence auditable, the appendix spells out the following protocol details:

- the exact baseline, Module B, and matched-capacity control architectures,
- how bidding-time and play-time public signals are implemented on the Doudizhu side,
- which tensors are available on the actor path at inference time,
- how the explicit-latent upper bound is implemented and why it violates C4,
- how public-channel corruption is performed in the intervention test,
- which statistics are reported: raw performance, normalized Δ_{coord} , and action-conditioned information estimates.

Three model classes. Within each domain, the C4-compliance study compares exactly three variants under matched architecture depth, width, replay settings, optimization hyperparameters, and training budget:

1. **No coordination:** the receiver uses no teammate-conditioned auxiliary objective.
2. **Explicit latent upper bound:** a sender-side latent is routed directly to the receiver action path at inference time. This model is intentionally illegal under C4 and is interpreted only as an empirical upper bound.
3. **Module B:** sender and receiver auxiliaries are present during training, but the receiver action path at inference time depends only on its local history and public-history features.

Per-domain base learners. Cross-domain heterogeneity is intentional. All Doudizhu-side variants are built on the same DMC-style per-position Q-network family, which keeps comparison to DouZero fair and exposes the $\max_a Q$ quantities consumed by Module C; the detailed budget / hardware / inherited-default disclosure relative to the public DouZero setup is collected in Appendix A.24. All futures-side variants are built on the same PPO-style actor-critic family, which is the natural fit for discrete execution templates and a centralized critic. The paper’s claim is therefore not that one base learner dominates the other, but that Modules A/B/C add the same qualitative capability on top of each domain’s appropriate base learner.

Shared Doudizhu replay recipe. All Doudizhu-side internal variants—Base, Base + Module A, Base + Module B, Base + Module C, and Full—use the same per-position prioritized replay throughout training. Priorities are updated from TD-style value error, sampling probabilities share one fixed exponent α , and importance-sampling correction shares one fixed exponent β ; warmup size, replay capacity, and priority floor are likewise held constant across variants. We treat this replay scheme as a shared training stabilizer for fair comparison against the internal ablations, not as a separate algorithmic module of the paper.

Compact cumulative attribution in the main paper. The Doudizhu ablation table in Section 6 uses a compact fixed-pool ablation against the Base DMC learner rather than a full round-robin: Base + Module A, Base + Module B, Base + Module A + Module B, and Full are each evaluated against that shared Base reference. This keeps the visible progression aligned with the critic-support / legal-coordination split while still showing a stand-alone Module-B row. The omitted stand-alone Base + Module C row appears below in the appendix text, and additional cumulative or side-by-side diagnostics beyond the compact table are treated only as supplementary evidence rather than as the primary reported result.

Doudizhu bidding model. The Doudizhu stack trains a dedicated bidding network in addition to the card-play networks. We keep bidding separate at the function-approximation level because the bid alphabet is small and scalar, whereas card-play actions are combinatorial; nevertheless, the two stages are unified at the information-structure level. The bidding model is conditioned only on private hand and public bid history, and its public outputs are inherited by the later farmer-play histories. When Module B is enabled, bidding-time and play-time heads are both trained under the same public-channel principle: public actions should expose coordination-relevant private strength without introducing any teammate-private execution-time channel. In the headline comparison to PerfectDou on the Doudizhu side, bidding is enabled on our model, and the primary summary statistic is the bidirectional average over seat assignments, because bidding is both part of the evaluation task and the first legal public-signaling stage used by Module B.

Public-channel intervention. The intervention test is run after training with all parameters frozen. For each evaluation episode, we create a corrupted copy of the receiver observation in which only the teammate-originating public-trace features are modified. In Doudizhu, these are the encoded teammate play-history channels; the receiver’s hand features, landlord-originated public actions, and legal-action mask are left unchanged. In the futures benchmark, these are the public order-flow features attributable to teammate execution behavior; the local order book, inventory, and action constraints remain untouched. We report

$$\Delta_{\text{pub}} = \text{Perf}(\text{normal}) - \text{Perf}(\text{corrupted public trace})$$

for each variant. A large Δ_{pub} for the C4-compliant A+B row together with a small Δ_{pub} for the matched-capacity control is interpreted as evidence that the gain flows through the legal public channel.

Evaluation-protocol scope. Table 1 uses a common seat-averaged tournament protocol over 10k randomly generated decks, so the main comparison is about coordination behavior under a matched evaluation script rather than about every possible external training budget. We use the replay-based execution benchmark in the same spirit: the point is not that it is a symmetric online adversary identical to Doudizhu, but that it exposes the same C4-limited public-trace coordination problem under a different action alphabet and a different base learner.

Cross-domain scope. The second domain therefore remains a structural stress test, not a claim that replay-based execution is a perfect game-theoretic mirror of Doudizhu. Its role is narrower and more falsifiable: if the same coordination ordering, recovery ratio, and public-trace corruption fragility reappear when bids/cards are replaced by orders/trades and DMC-style value learning is replaced by a PPO-style actor-critic, then the evidence is consistent with a shared AT-Dec-POSG bottleneck rather than a domain-specific artifact.

Figure summary. Table 4 carries the primary exact evidence for the matched C4-compliant A+B legal-coordination row, while Figure 2 below provides the same no-coordination / legal-coordination / explicit-latent ordering together with the Δ_{pub} intervention contrast in visual form. The training-curve and action-family figures in Appendix A.20 are the supporting analysis-side complement to Proposition 8: they show the within-domain progression over training and where the learned system appears to exploit the legal public channel most strongly.

A.20 Training-curve and action-level diagnostic figures

Reading Figure 1. These curves preserve the training-time version of the main ablation story. The useful point is not the exact transient shape of every line, but that the stronger A/B/C variants separate from the base early and keep that ordering across two structurally different domains, which supports the paper’s claim that the stack generalizes as an information-structure remedy rather than only as a domain-specific fit.

Reading Figure 2. This figure is the visual counterpart to Table 4. The bar ordering shows how much of the explicit-latent coordination benefit is recovered legally by the C4-compliant A+B row, and the side panel checks that this gain really flows through teammate-originating public traces rather than through an unrelated capacity increase.

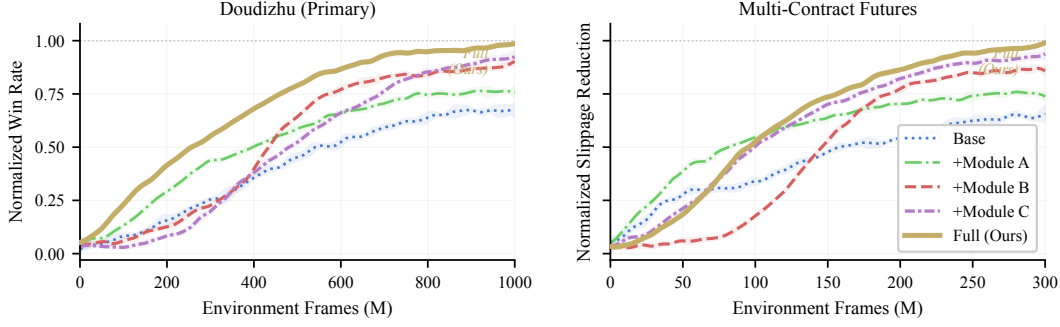


Figure 1: Within-domain normalized curves compare the canonical Base / +A / +B / +C / Full progression in Doudizhu and the matched replay-based stress-test variants in execution.

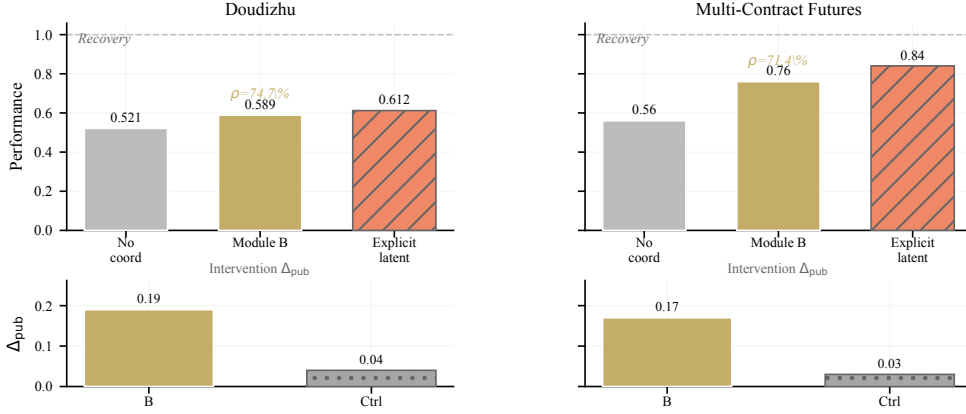


Figure 2: Cross-domain legal-coordination evidence chain. The main bars show the no-coordination / legal-coordination (A+B) / explicit-latent ordering that determines ρ_{coord} , while the side panel shows the public-trace intervention gap Δ_{pub} for the A+B row versus the matched-capacity control. The exact values appear in Table 4.

1292 **Reading Figure 3.** This diagnostic should be read as a probing result, not as a direct empirical
1293 estimate of the theorem-side mutual-information quantity. Its purpose is narrower: it indicates that
1294 the learned coordination signal concentrates in semantically consequential public actions such as
1295 bids, strong control plays, and aggressive execution moves, rather than in low-impact background
1296 behavior.

1297 A.21 Additional conclusion discussion

1298 The main text closes with a compressed conclusion. This subsection collects the longer interpretation
1299 text.

1300 **Cross-domain reading.** In the compact cumulative Doudizhu table, the distinctive evidence for
1301 Module B should be read from the Base→A→A+B progression together with the cross-domain
1302 recovery and intervention diagnostics, rather than from a stand-alone single row in isolation.

1303 **Action-family interpretation.** At the analysis level, the two domains also agree on where coordi-
1304 nation signal tends to concentrate: bids, high-commitment control plays, aggressive front-loads, and
1305 synchronized larger slices are more informative than background passes or passive low-participation
1306 actions. That pattern does not by itself prove Proposition 8, but it is consistent with the paper’s
1307 narrower claim that legal public actions can be structured as teammate-relevant signals without
1308 violating C4.

1309 **Practical-reading qualifier.** Once teammate-private state is excluded from execution, value support,
1310 legal signaling support, and adversarial coverage support stop being interchangeable engineering
1311 tricks and become distinct failure modes that should be diagnosed separately.

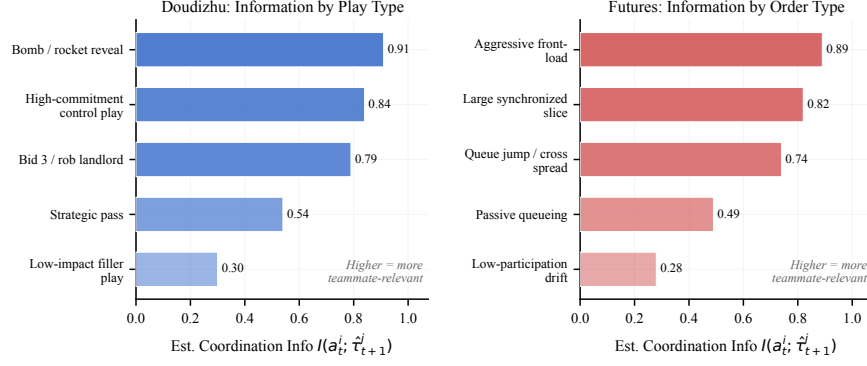


Figure 3: Estimated action-family information peaks at high-commitment public actions rather than low-impact background play.

Upper-bound and recovery statistics. The C4 upper-bound comparison reports four quantities per domain: the no-coordination baseline, the explicit-latent upper bound, the C4-compliant A+B score, and the coordination recovery ratio

$$\rho_{\text{coord}} = \frac{\text{Perf}(\text{A+B, C4-compliant}) - \text{Perf}(\text{no coordination})}{\text{Perf}(\text{explicit latent}) - \text{Perf}(\text{no coordination})}.$$

This ratio is preferred over raw deltas in the cross-domain analysis because it is dimensionless and therefore comparable across WP, ADP, shortfall, and risk-adjusted PnL.

A.22 Module C surrogate-ablation protocol

Because $\tilde{V}_{\text{farmer-joint}}$ is presented as a symmetry-based surrogate rather than a uniquely correct value, the appendix lists:

- the exact min/mean/max definitions,
- whether the same reward scale and optimizer settings are used across variants,
- the stability metric used for comparison,
- the threshold used in time-to-best-response reporting.

Joint-Q definition. The key extra step relative to the original two-player Minimax Exploiter is that the landlord’s opponent is a decentralized farmer pair rather than a single agent. We therefore define

$$\tilde{Q}_{\text{farmer-joint}}(s, a^{\text{up}}, a^{\text{down}}) = \frac{1}{2} (Q_F(o^{\text{up}}(s), a^{\text{up}}) + Q_F(o^{\text{down}}(s), a^{\text{down}})),$$

which induces the greedy team-side surrogate

$$\tilde{V}_{\text{farmer-joint}}(s) = \frac{1}{2} \left(\max_a Q_F(o^{\text{up}}(s), a) + \max_a Q_F(o^{\text{down}}(s), a) \right).$$

This choice is motivated by exchangeability of the two farmer roles under shared team reward and by the fact that execution remains decentralized under C4.

Exact surrogate variants. The ablation compares

$$\tilde{V}_{\text{farmer-joint}}^{\min}(s) = \min \left(\max_a Q_F(o^{\text{up}}(s), a), \max_a Q_F(o^{\text{down}}(s), a) \right),$$

$$\tilde{V}_{\text{farmer-joint}}^{\text{mean}}(s) = \frac{1}{2} \left(\max_a Q_F(o^{\text{up}}(s), a) + \max_a Q_F(o^{\text{down}}(s), a) \right),$$

$$\tilde{V}_{\text{farmer-joint}}^{\max}(s) = \max \left(\max_a Q_F(o^{\text{up}}(s), a), \max_a Q_F(o^{\text{down}}(s), a) \right),$$

with no other change to the league update rule or optimizer state. The mean surrogate is motivated by exchangeability of the two farmer roles under shared team reward; the ablation does not assume that the mean is uniquely correct.

1335 **Reporting targets.** Each surrogate is evaluated on four quantities: endpoint performance in
 1336 Doudizhu, endpoint performance in the futures benchmark, frames-to-threshold for exploiter improve-
 1337 ment, and a stability statistic such as seed variance or population robustness. The mean surrogate is
 1338 supported only if it attains a better overall tradeoff, rather than merely the best single endpoint metric.

1339 A.23 Non-positive shaping bound for Module C

1340 The original Minimax Exploiter notes that the opponent-value bonus should not create positive-reward
 1341 cycles. The same point carries over to the team setting once the opponent-side surrogates are shifted
 1342 into a nonnegative range.

1343 **Proposition 11** (Non-positive shifted minimax bonuses). *Let m_F and m_L be lower bounds, or*
 1344 *running lower-bound estimates, such that*

$$\bar{V}_{\text{farmer-joint}}(s) = \tilde{V}_{\text{farmer-joint}}(s) - m_F \geq 0, \quad \bar{V}_L(s) = \max_a Q_L(o^L(s), a) - m_L \geq 0.$$

1345 *Define the shifted shaping bonuses*

$$B_t^{\text{land-exp}} = -\alpha\gamma(1 - d_t) \bar{V}_{\text{farmer-joint}}(s_{t+1}), \quad B_t^{\text{farm-exp}} = -\alpha\gamma(1 - d_t) \bar{V}_L(s_{t+1}),$$

1346 *with $\alpha \geq 0$, $\gamma \in [0, 1]$, and $d_t \in \{0, 1\}$. Then*

$$B_t^{\text{land-exp}} \leq 0, \quad B_t^{\text{farm-exp}} \leq 0.$$

1347 **Proof.** Both shifted value surrogates are nonnegative by construction, and α , γ , and $(1 - d_t)$ are
 1348 also nonnegative. Each bonus is therefore the negative of a nonnegative product and is at most zero.

1349 A.24 Compute resources, asset provenance, and release plan

1350 **Compute reporting.** For the Doudizhu-side benchmark, the reported settings are: one machine with
 1351 64 Intel Xeon Silver 4214R CPU cores and learner hardware = $4 \times$ RTX 4090 GPUs; training budget
 1352 = 1000M episodes per reported internal run; evaluation protocol = bidding-enabled seat-averaged
 1353 10k-deck matches against reproduced DouZero / PerfectDou opponents. Additional reproducibility
 1354 fields include actor processor type, learner GPU type, available accelerator memory, per-run wall-
 1355 clock time, and total compute across the reported seeds.

1356 **Parity statement.** The paper therefore claims a matched evaluation protocol, not fully matched
 1357 training compute. Our internal variants are trained under the reported $4 \times$ RTX 4090 / 1000M-episode
 1358 budget, while the public DouZero / PerfectDou baselines are reproduced and re-evaluated under
 1359 the shared bidding-enabled 10k-deck script rather than retrained from scratch under a new uniform
 1360 compute budget. This keeps the protocol comparison auditable without overstating compute parity.

1361 **Existing assets and licenses.** The Doudizhu-side evaluation protocol builds directly on the public
 1362 assets released alongside DouZero and PerfectDou [23, 28]. The public DouZero release identifies
 1363 Apache-2.0 terms, and the public PerfectDou release exposes pretrained models and evaluation code
 1364 while noting that the original distributed training code is not fully released. In this submission, reuse
 1365 is documented at the level of environment conventions, evaluation scripts, public checkpoints, and
 1366 reproduced baselines, with the cited upstream license terms retained where applicable.

1367 **Finance-side data access.** The futures benchmark separates the replay engine from the market
 1368 data used to drive it. When the benchmark is instantiated on redistributable public or academic
 1369 data, the relevant source, version, and terms of use are the identifying metadata for reproducibility.
 1370 When it relies on licensed historical market data that cannot be redistributed, the reproducibility
 1371 path is the non-restricted pipeline: preprocessing code, feature schema, replay engine, event-window
 1372 identifiers or hashes where legally permissible, action templates, and evaluation scripts. Under that
 1373 restricted-data regime, reproducibility rests on an auditable experimental pipeline rather than on raw
 1374 data release.

New assets and documentation. The benchmark documentation fields relevant to this paper are: task definition, contract universe, train/validation/test split logic, preprocessing steps, action templates, reward definition, environment API, and exact commands for reproducing the main tables and figures. In the present anonymized submission, these fields define the intended reproducibility schema rather than a claim that every finance-side field value is already bundled in the paper package. The accompanying release note distinguishes which results are reproducible from open assets and which rely on restricted data access. This separation keeps the structural contribution of the second domain auditable even when the raw data source itself cannot be shipped openly.

1383 A.25 Additional Module-A implementation details

Implementation view. The main-text equation should be read as a schematic CTDE-level description of the critic-side information made available during Module-A training, while this appendix gives the concrete per-domain implementation that realizes that schema via a local student value path and a training-only privileged teacher branch. The learner instantiates Module A with a local student value path plus a training-only privileged teacher branch. The student critic consumes only features derived from $\tau_t^i = (h_t^{\text{pub}}, x_t^i)$, while the teacher branch encodes s_t^{full} and regresses the same return target under privileged state. The distillation term then pulls the student value prediction toward the teacher value prediction without exposing privileged tensors to the execution policy. In the Doudizhu instantiation, the belief heads are attached to the DMC-style per-position value stack and predict the hidden hands of the two unseen players; in the futures instantiation, they augment the PPO critic and predict latent execution-state quantities such as hidden liquidity and pressure. The resulting belief predictions are re-encoded and fed back into the student value path, so the critic is not merely supervised to guess hidden state, but also uses that inferred hidden-state summary when estimating value. In the PPO-style futures instantiation, actor advantages are computed from the student value path rather than the privileged teacher branch; the teacher is used only to supply the distillation target during training. Execution-time action selection remains unchanged: privileged features, teacher outputs, and belief targets are all absent from the actor path.

Mechanical defaults used in the released code paths. For the replay-based execution benchmark, the released PPO configuration uses $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$, clip $\epsilon = 0.20$, learning rate 3×10^{-4} , entropy coefficient 0.01, value coefficient 0.50, hidden width 128, and 4 update epochs. The Module-A auxiliary enters that multitask objective with task weight 0.25. On the Doudizhu side, the matching DMC code path exposes `teacher_value_coef= 0.5`, `distill_coef= 0.25`, and `belief_coef= 0.05` for the privileged teacher and hidden-hand belief terms.

1407 A.26 Additional Module-B implementation details

Lemma 1 implies that decentralized teams suffer a structural coordination barrier whenever optimal coordinated behavior depends on teammate-private state. Standard private communication channels would violate C4, so the only legal alternative is the public action history.

Across domains, the semantics differ but the structure is shared: in Doudizhu, Module B begins with the low-bandwidth public bid history inherited by the realized farmer continuation game and continues through card play; in futures execution, it is expressed through execution cadence or urgency. In our Doudizhu implementation, bidding uses a separate network but the same C4-compliant public channel. This is a core part of the mechanism rather than bookkeeping: the bid sequence is legal public pre-history that can expose hand-strength information to the realized farmer team before the first card is played.

We therefore use an implicit coordination mechanism based on auxiliary supervision: training encourages action-side representations to encode coordination-relevant private state, while execution keeps the same legal policy signature. Let $z_t^i = g_{\theta_i}(\tau_t^i)$ denote sender-side features and let $\hat{z}_{t+1}^{i \rightarrow j} = h_{\psi_j}(o_{1:t+1}^{\text{pub}}, \tau_t^j)$ denote receiver-side reconstruction from public observations only.

Per-domain targets and losses. The shared template uses sender and receiver auxiliary classifiers

$$\begin{aligned}\mathcal{L}_{\text{sender}} &= \text{CE}(\text{coord_sender_logits}, y_{\text{sender}}^i), \\ \mathcal{L}_{\text{receiver}} &= \text{CE}(\text{coord_receiver_logits}, y_{\text{receiver}}^{i \rightarrow j}).\end{aligned}$$

1423 In Doudizhu, y_{sender}^i summarizes private hand-strength / remaining-hand categories and $y_{\text{receiver}}^{i \rightarrow j}$
 1424 asks whether the teammate can reconstruct that summary from the bid/play trace. In multi-contract
 1425 execution, we instantiate the same template directly from the replay environment. Let

$$u_t^i = \text{clip}\left(\frac{|r_t^i|}{\max(|r_0^i|, \epsilon_r)} \cdot \frac{T}{\max(T - t, 1)}, 0, 1\right), \quad \epsilon_r > 0,$$

1426 be the hidden urgency target for contract i , where r_t^i is remaining inventory and T is the episode
 1427 horizon. Then $y_{\text{sender}}^i = \text{bin}(u_t^i)$, and the receiver target for agent j is $y_{\text{receiver}}^j = \text{bin}(\bar{u}_t^{-j})$ with
 1428 $\bar{u}_t^{-j} = \frac{1}{N-1} \sum_{i \neq j} u_t^i$. The receiver is supervised against that hidden teammate-mean target while
 1429 consuming only its own local history and the legal public tape, so the target itself never enters
 1430 execution-time inference. In the futures implementation, these cross-entropy heads are paired with
 1431 the same public-action latent-alignment term used in the shared Module-B template. Their role is to
 1432 shape representations densely during training; the actual behavioral shift toward informative legal
 1433 public actions is still mediated by the shared RL objective.

1434 **Mechanical defaults used in the released code paths.** In the replay-based execution benchmark,
 1435 Module B uses task weight 0.10, latent-alignment weight 0.05, coordination latent width 32, and
 1436 sender/receiver urgency bins with thresholds $[0.20, 0.45, 0.70]$, producing four urgency classes. On
 1437 the Doudizhu side, the corresponding released supervision weights are `coord_sender_coef` = 0.05
 1438 and `coord_receiver_coef` = 0.05. These constants are fixed paper-side defaults rather than hidden
 1439 sweep parameters.

1440 **C4 compliance.** Training may shape informativeness, but execution remains

$$\pi_i(a \mid \tau_t^i),$$

1441 with no private latent channel, whether the realized action law is stochastic as in the PPO-style
 1442 execution benchmark or degenerate at a greedy action as in the DMC-style Doudizhu stack. Formally,
 1443 farmer j 's action is measurable with respect to $\sigma(\tau_t^j)$, and any information about farmer i enters only
 1444 through the legal public history already contained in τ_t^j .

1445 **Explicit-latent upper bound.** The explicit-latent upper bound is a deliberately C4-violating di-
 1446 agnostic architecture used only to estimate how much coordination value remains above the legal
 1447 Module-B variant under the same task and evaluation script. Concretely, it injects a sender-side
 1448 private coordination latent into the receiver's action path at inference time: in Doudizhu, the la-
 1449 tent summarizes the sender hand-strength / remaining-hand target, and in replay-based execution
 1450 it summarizes the sender urgency bin or its continuous precursor; in both domains, that latent is
 1451 concatenated to the receiver-side action-head features before the final policy logits are produced. This
 1452 intervention violates C4 because the deployed receiver action is no longer measurable with respect to
 1453 $\sigma(\tau_t^j)$: it depends on teammate-private state routed through a non-public latent tensor. We therefore
 1454 use the explicit-latent model only as a diagnostic upper bound on recoverable coordination benefit,
 1455 not as an admissible baseline or deployable policy.

1456 **Contrast with explicit communication.** We contrast the legal variant with this C4-violating
 1457 explicit-latent ablation because it makes the recoverable coordination headroom visible. Once
 1458 teammate-side latent tensors are removed, the executed policy still remains $a_t^i \sim \pi_i(\cdot \mid \tau_t^i)$, unlike
 1459 explicit-communication architectures such as CommNet, RIAL/DIAL, or TarMAC.

1460 A.27 Additional Module-C implementation details

1461 **General league structure.** Adversarial team training is prone to cycling, so we maintain three
 1462 policy populations: main policies for robustness, snapshot policies for diversity, and exploiter policies
 1463 for approximate best response.

1464 The generic league template is the same across domains. Main policies are the robustness anchor
 1465 used to define the current target behavior. Snapshot policies retain earlier strategies and therefore
 1466 broaden the opponent pool. Exploiter policies are trained as approximate best responses against
 1467 the current main side so that newly exposed weaknesses can be reinserted into the league. This
 1468 main/snapshot/exploiter split is the population-level structure of Module C; the domain-specific

question is how the exploiter reward should use opponent-side value information. In replay-based execution, the league pool is a finite set of tagged market regimes rather than a set of learned market-side policies: the implementation samples from `main_balanced`, `snapshot_wide_spread`, `snapshot_fast_tape`, and `exploiter_stressed_liquidity`. Rollouts are drawn with the configured main/snapshot/exploiter probabilities, and only exploiter-tagged scenarios activate the non-positive shifted minimax bonus. Under this reading, the execution-side league is a mechanical robustness curriculum over replay scenarios, not a hedge that hides missing opponent learning.

Doudizhu instantiation. Relative to the original two-player Minimax Exploiter [1], the extra issue in Doudizhu is that the landlord does not face a single opponent policy with a single Q-interface. Instead, the opponent is a decentralized two-farmer team with shared reward and separate private observations. The first step is therefore to define a team-side opponent value.

For the two-farmer case, let the two farmer critics be evaluated from their own local observations. We define the symmetric joint surrogate

$$\tilde{Q}_{\text{farmer-joint}}(s, a^{\text{up}}, a^{\text{down}}) = \frac{1}{2} (Q_F(o^{\text{up}}(s), a^{\text{up}}) + Q_F(o^{\text{down}}(s), a^{\text{down}})).$$

Since the two farmers act independently at execution time, the induced greedy team-side value is

$$\tilde{V}_{\text{farmer-joint}}(s) = \frac{1}{2} \left(\max_a Q_F(o^{\text{up}}(s), a) + \max_a Q_F(o^{\text{down}}(s), a) \right).$$

The mean is the simplest symmetry-preserving aggregation of the two farmer critics under shared team reward, and is the minimal extra definition needed to lift Minimax Exploiter from a two-player zero-sum game to an asymmetric 2-vs-1 AT-Dec-POSG. Both reward formulas keep the same negative sign because the inserted quantity is always the opponent-side value proxy in that opponent’s own reward coordinates. Thus, a farmer exploiter still improves by driving the landlord critic downward.

Non-positive shifted bonus. To match the original reward-hacking safeguard, let

$$\tilde{V}_{\text{farmer-joint}}(s) = \tilde{V}_{\text{farmer-joint}}(s) - m_F \geq 0, \quad \bar{V}_L(s) = \max_a Q_L(o^L(s), a) - m_L \geq 0,$$

where m_F and m_L are lower bounds, or running lower-bound estimates, for the corresponding opponent-side value proxies. The shifted bonuses then become

$$B_t^{\text{land-exp}} = -\alpha\gamma(1 - d_t) \tilde{V}_{\text{farmer-joint}}(s_{t+1}), \quad B_t^{\text{farm-exp}} = -\alpha\gamma(1 - d_t) \bar{V}_L(s_{t+1}).$$

Since $\alpha \geq 0$, $\gamma \in [0, 1]$, $(1 - d_t) \geq 0$, and both shifted surrogates are nonnegative, each bonus is at most zero. So the exploiter still sees opponent-aware shaping pressure, but the extra term cannot itself generate unbounded positive cycles.

Mechanical defaults used in the released code paths. For the replay-based execution benchmark, the released scenario league contains one main regime (`main_balanced`), two snapshot regimes (`snapshot_wide_spread`, `snapshot_fast_tape`), and one stressed-liquidity scenario family (`exploiter_stressed_liquidity`). Sampling probabilities are 0.5/0.3/0.2 for main/snapshot/exploiter roles, with uniform sampling within each role. The stressed-liquidity scenario family is refreshed every 50 episodes, and the shifted-bonus parameters are $\alpha = 0.12$, $\gamma = 1.0$, and value floor $m_F = m_L = 0$. On the Doudizhu side, the released league keeps 6 snapshots and 4 exploiters per group, uses opponent-sampling probabilities 0.2/0.5/0.3 for main/snapshot/exploiter sources, trains an exploiter rather than a main policy with probability 0.33, refreshes opponent-pool metadata every 20 episodes, and resets exploiters from the live main policy every 20,000 frames.

A.28 Additional experimental details and supplementary tables

This subsection collects the longer secondary experiment material used to support the compact summaries in Section 6.

Experimental questions. The empirical program is organized around four checks. First, the headline tables contain the complete main-domain and second-domain comparisons, because the paper’s structural argument is not meaningful if the main evidence is incomplete. Second, the within-domain ablations show whether Module A, B, and C contribute in the predicted order. Third, the intervention protocol tests whether corrupting teammate-originating public traces specifically hurts the C4-compliant coordinated policy. Fourth, the second domain reproduces the same ordering and recovery pattern without requiring the raw metric scale to match Doudizhu.

1514 **Main-text protocol matrix.**

Artifact	Domain	Comparison pool	Evaluation script
Table 1	Doudizhu	reproduced AT-Dec-POSG / PerfectDou / DouZero-WP / DouZero-ADP	seat-averaged 10k-deck tournament with bidding enabled
Table 2	Doudizhu	fixed Base DMC opponent pool	shared ablation script, 3 matched seeds, fixed-pool comparison rather than round-robin
Table 3	replay-based execution	PPO executor versus follow-up checkpoints	matched replay benchmark, 3 matched seeds
Tables 4–5	cross-domain A+B diagnostics	no-coord. / A+B / explicit-latent / control variants	matched C4-compliance scripts and held-out probe protocol

1516 **Action-family probing formula.** For the action-level signal-structure analysis, we additionally
 1517 bucket public actions by semantic type and estimate

$$\hat{I}(a; \hat{z}^{i \rightarrow j}) = \hat{H}(a) - \hat{H}(a \mid \hat{z}^{i \rightarrow j}),$$

1518 where $\hat{z}^{i \rightarrow j}$ is the receiver-side reconstruction of sender intent produced by Module B. The entropies
 1519 are estimated by Monte Carlo counts on held-out trajectories; for Doudizhu, the buckets include bids
 1520 and card-play families. This quantity is a secondary diagnostic for where signal concentrates in the
 1521 public action alphabet. The primary capacity probe in the main paper is the lower-bound quantity in
 1522 Appendix A.14; $\hat{I}(a; \hat{z})$ should be read only as an action-family complement rather than as a direct
 1523 estimate of theorem-side channel capacity.

1524 **Module-B evidence chain.** We do not ask any single statistic to carry the entire coordination
 1525 claim. Instead, Module B is evaluated with a triangular evidence chain: the legal-capacity lower
 1526 bound $H(Z_t \mid x_t^j, C_t) - \mathcal{L}_{\text{rec}}^{\text{held}}$ together with its no-public control gap, the gap to a C4-violating
 1527 explicit-latent upper bound, and the public-trace corruption intervention. In the matched cross-domain
 1528 comparison, the deployed C4-compliant row is Base + Module A + Module B in both domains.
 1529 The lower bound asks how decodable sender-relevant summaries become from the legal public trace,
 1530 the explicit-latent upper bound quantifies the coordination value still left on the table under C4, and
 1531 the corruption test asks whether the deployed gain actually depends on teammate-originating public
 1532 traces at evaluation time. The auxiliary $\hat{I}(a; \hat{z}^{i \rightarrow j})$ diagnostic remains useful only for localizing
 1533 which action families appear most signal-bearing.

1534 **Supplementary omitted Doudizhu row.** The compact Doudizhu table in Section 6 omits one
 1535 stand-alone row to save space: under the same league-trained, evaluated-against-Base protocol, the
 1536 stand-alone Base + Module C variant reaches $\text{WP} = 59.30\%$ and $\text{ADP} = 0.63$, corresponding to
 1537 $+7.20$ WP and $+0.32$ ADP over Base.

1538 **Public-channel intervention.** The intervention test is run after training with all parameters frozen.
 1539 For each evaluation episode, we create a corrupted copy of the receiver observation in which only
 1540 the teammate-originating public-trace features are modified. In Doudizhu, these are the encoded
 1541 teammate play-history channels; the receiver’s hand features, landlord-originated public actions,
 1542 and legal-action mask are left unchanged. In the futures benchmark, these are the public order-flow
 1543 features attributable to teammate execution behavior; the local order book, inventory, and action
 1544 constraints remain untouched. We report

$$\Delta_{\text{pub}} = \text{Perf}(\text{normal}) - \text{Perf}(\text{corrupted public trace})$$

1545 for each variant. A large Δ_{pub} for the C4-compliant A+B row together with a small Δ_{pub} for
 1546 the matched-capacity control is interpreted as evidence that the gain flows through the legal public
 1547 channel.

1548 **Fragility reading.** This same intervention statistic is also a practical fragility diagnostic: a larger
 1549 public-trace corruption gap is desirable as evidence that the legal public channel is genuinely being
 1550 used, but it also means the learned policy is more exposed to public-trace corruption. We therefore
 1551 read this quantity as mechanism evidence first and robustness risk second, and leave corruption-aware
 1552 mitigation to future work rather than claim it is already solved here.

Diagnostic interpretation. The empirical evidence for the legal-coordination component now comes from Table 4: the large Δ_{pub} gaps for the A+B row relative to the matched-capacity controls indicate that teammate-originating public traces are being used as coordination carriers in both domains. Figure 2 provides the same ordering as a supplementary visual summary, while the action-family probing figure in Appendix A.20 should still be read only as a representation-level proxy rather than as a direct empirical estimate of the theorem-side quantity $I(a_t^i; o_{t+1}^j)$.

A.29 Cross-domain protocol note and supplementary visual summary

Table 4 carries the compact cross-domain evidence chain. This appendix subsection gives the longer protocol note and the supplementary visual summary.

Protocol note. For the Doudizhu row, all reported variants are evaluated against the same fixed Base DMC opponent under the same seat-averaged protocol over 10k decks; the no-coordination baseline is that Base learner, and the C4-compliant coordination row corresponds to Base + Module A + Module B. The execution row uses the matched replay-based stress-test protocol, where Perf is the normalized larger-is-better slippage-reduction score rather than raw shortfall.

Cross-domain metric definitions. The naming convention is fixed: Δ_{coord} is the total explicit-minus-no-coordination gap, Δ_{unrec} is the explicit-minus-legal unrecovered residue, and the two are never swapped:

$$\begin{aligned}\Delta_{\text{coord}} &= \text{Perf}(\text{explicit coordination, violates C4}) - \text{Perf}(\text{no coordination}), \\ \Delta_{\text{unrec}} &= \text{Perf}(\text{explicit coordination, violates C4}) - \text{Perf}(\text{A+B, C4-compliant}), \\ \rho_{\text{coord}} &= \frac{\text{Perf}(\text{A+B, C4-compliant}) - \text{Perf}(\text{no coordination})}{\text{Perf}(\text{explicit coordination, violates C4}) - \text{Perf}(\text{no coordination})} = 1 - \frac{\Delta_{\text{unrec}}}{\Delta_{\text{coord}}}.\end{aligned}$$

Here Perf denotes a within-domain larger-is-better scalar used only for the C4-compliance protocol: in Doudizhu it is seat-averaged win rate against the fixed opponent pool, and in replay-based execution it is the normalized slippage-reduction score under the matched stress-test script.

Reading Figure 2. Figure 2 should be read as the visual complement to Table 4 rather than as the primary headline artifact. The replay-based execution benchmark is structurally more heterogeneous than the card-game benchmark in its action alphabet, horizon pressure, and market-regime variation, yet the same no-coordination < A+B < explicit-latent ordering still appears immediately under the matched evaluation scripts. Read together with the Base → A → A+B → Full ladder, this persistence supports the intended qualitative claim for the full A/B/C stack: once critic support, legal public-channel coordination, and league robustness are switched on, the method exhibits transferable coordination structure early rather than requiring domain-specific re-interpretation of the bottleneck.

A.30 Execution Pareto table

Table 8: Replay-based multi-contract execution Pareto sweep. The Full row in Section 6 corresponds to the selected validation-qualified, completion-constrained follow-up; the earlier incumbent and the additional follow-up points are shown here for comparison. Here Cost is the negative of the shortfall-based team reward, so smaller values mean lower total execution cost.

Variant	Cost	Shortfall	RA-PnL	Comp. rate	Trace-frag.
Primary Full (main text)	3.8948	3.2909	-3.3564	0.9772	0.2560
Full incumbent	3.9310	3.1631	-3.2464	0.9568	0.2550
completion_005	5.2360	4.7288	-4.7808	0.9875	0.5555
trace_consistency_002	5.6870	4.1705	-4.2840	0.6645	0.1701

Reading Table 8. On the Replay-Based Multi-Contract Execution Stress Test, the Full checkpoint reported in Section 6 is the validation-qualified, completion-constrained follow-up: among follow-up checkpoints whose completion meets or exceeds the PPO reference, we report the lowest-Cost point. This selected Full checkpoint still clearly beats PPO on Cost (3.8948 vs. 6.3985), shortfall (3.2909 vs. 5.7538), and risk-adjusted PnL (-3.3564 vs. -5.8130), while also restoring completion to 0.9772, which exceeds PPO’s 0.9524. The earlier Full incumbent is retained as a nearby Pareto comparison

with slightly better shortfall and risk-adjusted PnL but lower completion (0.9568). We therefore treat the incumbent and the remaining sweep entries as supplementary trade-off points rather than as the primary headline result.

A.31 Supplementary surrogate-ablation table

The main paper keeps only the short surrogate-ablation conclusion. This appendix subsection gives the fuller reading, the figure, and the compact min/mean/max table.

Surrogate-ablation reading. Module C uses a symmetry-preserving mean surrogate rather than treating min/max aggregation as axiomatic alternatives. Figure 4 shows that mean gives the best overall endpoint/speed/stability tradeoff across the two domains, so it is used as the default team-side exploiter value proxy.

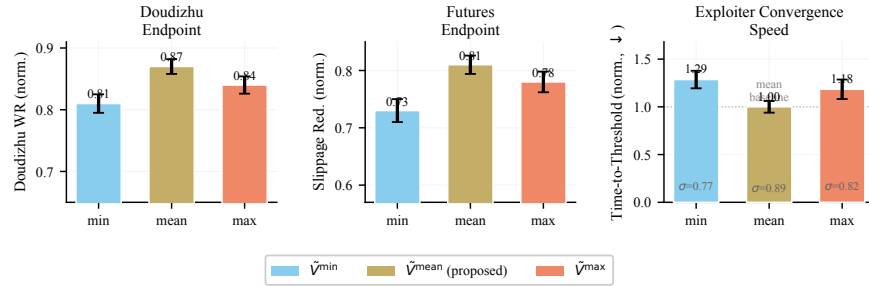


Figure 4: Module C surrogate ablation: mean gives the strongest overall endpoint/speed/stability tradeoff.

Reading Figure 4. The point of this figure is comparative rather than absolute. It visualizes why the paper keeps the mean surrogate as the default Module-C choice: among the three symmetric aggregations, it is the most balanced option across final performance, convergence speed, and held-out robustness, without relying on a single optimistic or pessimistic extreme.

Table 9: Surrogate ablation across Doudizhu and replay-based multi-contract execution. ‘Time’ reports millions of environment frames to reach the fixed exploiter-improvement threshold, and ‘Stab.’ reports the normalized held-out population-robustness score in $[0, 1]$, where larger is better.

Agg.	DZ	Exec.	Time↓	Stab.
min	0.81	0.73	63.0	0.77
mean	0.87	0.81	49.0	0.89
max	0.84	0.78	58.0	0.82

Reading Table 9. The table provides the exact values behind Figure 4. The mean aggregation is the only variant that is simultaneously strongest or near-strongest on both domains, fastest to threshold, and best on the stability score, which is why it is treated as the default surrogate rather than as an arbitrary aesthetic choice.

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Justification: The abstract and introduction explicitly frame AT-Dec-POSG as a constrained subclass of POSG/ATMG, describe Module B as C4-compliant, and restrict the empirical claim to cross-domain evidence rather than universal generalization; see the abstract and Sections 1 and 6.

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